



Cloud Computing

BSE-VB

Submitted By

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Submitted to

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LAB-06

Lab 6 – Linux Users, Groups, Permissions, Pipes, and Bash Scripting

Task 1 – Switch to root with su - and back to a normal user

1. Set a root password so you can use su -.

Screenshot: task1_set_root_password.png

```
tooba65@ubuntu:~$ sudo passwd root
[sudo] password for tooba65:
Sorry, try again.
[sudo] password for tooba65:
New password:
Retype new password:
passwd: password updated successfully
```

2. Switch to the root account and verify using whoami and id.

Screenshot: task1_su_root.png

```
tooba65@ubuntu:~$ su -
Password:
root@ubuntu:~# whoami
root
root@ubuntu:~# id
uid=0(root) gid=0(root) groups=0(root)
root@ubuntu:~#
```

3. Exit from root and return to your normal user account.

Screenshot: task1_exit_to_user.png

```
root@ubuntu:~# exit
logout
tooba65@ubuntu:~$ whoami
tooba65
tooba65@ubuntu:~$
```

Task 2 – Create user tom and verify in passwd/group/shadow

1. Create a new user named **tom**.

Screenshot: task2_adduser_tom.png

```
tooba65@ubuntu:~$ sudo adduser tom
info: Adding user `tom' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `tom' (1001) ...
info: Adding new user `tom' (1001) with group `tom (1001)' ...
info: Creating home directory `/home/tom' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for tom
Enter the new value, or press ENTER for the default
    Full Name []: tom
    Room Number []: 3
    Work Phone []: 0322
    Home Phone []: 2
    Other []: 1
Is the information correct? [Y/n] y
info: Adding new user `tom' to supplemental / extra groups `users' ...
info: Adding user `tom' to group `users' ...
tooba65@ubuntu:~$
```

2. Verify that tom appears in **/etc/passwd**.

Screenshot: task2_verify_passwd.png

```
tooba65@ubuntu:~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534:/:nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
systemd-timesync:x:997:997:systemd Time Synchronization:/:/usr/sbin/nologin
```

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3. Verify that tom appears in **/etc/group**.

Screenshot: task2_verify_group.png

```
pulse:x:122:
pulse-access:x:123:
xrdp:x:124:
docker:x:988:
tom:x:1001:
```

4. Verify that tom appears in **/etc/shadow**.

Screenshot: task2_verify_shadow.png

```
tooba65@ubuntu:~$ sudo cat /etc/shadow
root:$y$j9T$5n11Q4WnfucTDgcwApuFM0$oDpPhbg8HgzB58FcTo/2zoMDV15jszdk0Gwps6rUU8:20397:0:99999:7:::
daemon*:20305:0:99999:7:::
bin*:20305:0:99999:7:::
sys*:20305:0:99999:7:::
sync*:20305:0:99999:7:::
games*:20305:0:99999:7:::
man*:20305:0:99999:7:::
lp*:20305:0:99999:7:::
mail*:20305:0:99999:7:::
news*:20305:0:99999:7:::
uucp*:20305:0:99999:7:::
proxy*:20305:0:99999:7:::
www-data*:20305:0:99999:7:::
backup*:20305:0:99999:7:::
list*:20305:0:99999:7:::
irc*:20305:0:99999:7:::
_apt*:20305:0:99999:7:::
nobody*:20305:0:99999:7:::
systemd-network:!*:20305:0:0:0:
systemd-timesync:!*:20305:0:0:0:
dhcpcd:!:20305:0:0:0:
messagebus:!:20305:0:0:0:
systemd-resolve:!*:20305:0:0:0:
pollinate:!:20305:0:0:0:
polkitd:!*:20305:0:0:0:
syslog:!:20305:0:0:0:
uidd:!:20305:0:0:0:
tcpdump:!:20305:0:0:0:
tss:!:20305:0:0:0:
landscape:!:20305:0:0:0:
fwupd-refresh:!*:20305:0:0:0:
```

Task 3 – Create groups; change tom's primary and secondary groups

1. Create groups **developer**, **devops**, and **designer** and verify in **/etc/group**.

Screenshot: task3_groupadd.png

```
tooba65@ubuntu:~$ cat /etc/group
tom:x:1001:
developer:x:1002:
devops:x:1003:
designer:x:1004:
```

2. Change tom's **primary group** to designer and verify with id.

Screenshot: task3_change_primary_group.png

```
tooba65@ubuntu:~$ sudo usermod -g designer tom
tooba65@ubuntu:~$ id tom
uid=1001(tom) gid=1004(designer) groups=1004(designer),100(users)
```

3. Add tom to **secondary groups** developer and devops, then verify.

Screenshot: task3_add_secondary_groups.png

```
tooba65@ubuntu:~$ sudo usermod -aG developer,devops tom
tooba65@ubuntu:~$ groups tom
tom : designer users developer devops
```

4. Reset tom's secondary groups to default (only tom's own group) and verify.

Screenshot: task3_reset_secondary_groups.png

```
tooba65@ubuntu:~$ sudo usermod -G tom tom
tooba65@ubuntu:~$ id tom
uid=1001(tom) gid=1004(designer) groups=1004(designer),1001(tom)
tooba65@ubuntu:~$ groups tom
tom : designer tom
tooba65@ubuntu:~$
```

Task 4 – Create/delete users (Jerry, Scooby) and groups (jolly, anime)

1. Create users **Jerry** and **Scooby**.

Screenshot: task4_add_users.png

```

tooba65@ubuntu:~$ sudo adduser Jerry
err: Please enter a username matching the regular expression
      configured via the NAME_REGEX configuration variable. Use the
      '--allow-bad-names' option to relax this check or reconfigure
      NAME_REGEX in configuration.
tooba65@ubuntu:~$ sudo adduser jerry
info: Adding user `jerry' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `jerry' (1005) ...
info: Adding new user `jerry' (1005) with group `jerry (1005)' ...
info: Creating home directory `/home/jerry' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for jerry
Enter the new value, or press ENTER for the default
    Full Name []: jerry
    Room Number []: 4
    Work Phone []: 03222
    Home Phone []: 03444
    Other []: 1
Is the information correct? [Y/n] y
info: Adding new user `jerry' to supplemental / extra groups `users' ...
info: Adding user `jerry' to group `users' ...
tooba65@ubuntu:~$

```

2. Try logging in as Scooby (it should fail since there's no password).

Screenshot: task4_scooby_su_auth_failure.png

```

tooba65@ubuntu:~$ sudo useradd Scooby
tooba65@ubuntu:~$ su - Scooby
Password:
su: Authentication failure
tooba65@ubuntu:~$

```

3. Set a password for Scooby.

Screenshot: task4_set_password_scooby.png

```

tooba65@ubuntu:~$ sudo passwd Scooby
New password:
Retype new password:
passwd: password updated successfully
tooba65@ubuntu:~$

```

4. Log in as Scooby again (home directory missing warning appears).

Screenshot: task4_scooby_su_no_home.png

```

tooba65@ubuntu:~$ su - Scooby
Password:
su: warning: cannot change directory to /home/Scooby: No such file or directory
$ exit
tooba65@ubuntu:~$

```

5. Verify Scooby's missing home directory.

Screenshot: task4_scooby_no_home.png

```
tooba65@ubuntu:~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534:./nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:./usr/sbin/nologin
systemd-timesync:x:997:997:systemd Time Synchronization:./usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,./usr/lib/dhcpcd:/bin/false
messagebus:x:101:102:./nonexistent:/usr/sbin/nologin
systemd-resolve:x:992:992:systemd Resolver:./usr/sbin/nologin
pollinate:x:102:1:./var/cache/pollinate:/bin/false
polkitd:x:991:991:User for polkitd:./usr/sbin/nologin
syslog:x:103:104:./nonexistent:/usr/sbin/nologin
uidd:x:104:105:./run/uidd:/usr/sbin/nologin
tcpdump:x:105:107:./nonexistent:/usr/sbin/nologin
tss:x:106:108:TPM software stack,,./var/lib/tpm:/bin/false
```

```
tooba65@ubuntu:~$ ls -ld /home/Scooby
ls: cannot access '/home/Scooby': No such file or directory
tooba65@ubuntu:~$
```

6. Create Scooby's home directory manually and set ownership.

Screenshot: task4_scooby_create_home.png

```
tooba65@ubuntu:~$ sudo mkdir -p /home/Scooby
tooba65@ubuntu:~$ sudo chown Scooby:Scooby /home/Scooby
tooba65@ubuntu:~$ sudo chmod 750 /home/Scoo
chmod: cannot access '/home/Scoo': No such file or directory
tooba65@ubuntu:~$ sudo chmod 750 /home/Scooby
tooba65@ubuntu:~$ ls -ld /home/Scooby
drwxr-x--- 2 Scooby Scooby 4096 Nov  5 13:43 /home/Scooby
tooba65@ubuntu:~$
```

7. Log in as Scooby again and verify successful login.

Screenshot: task4_scooby_login_success.png

```

tooba65@ubuntu:~$ su - Scooby
Password:
$ pwd
/home/Scooby
$ ls -la
total 8
drwxr-x--- 2 Scooby Scooby 4096 Nov  5 13:43 .
drwxr-xr-x 6 root   root   4096 Nov  5 13:43 ..
$

```

8. Verify users in **/etc/passwd** and note Scooby's shell.

Screenshot: task4_verify_users.png

```

nloop31c:x:113:117::/nonexistent:/bin/rdisc
avahi:x:114:118:Avahi mDNS daemon,,,:/run/avahi-daemon:/usr/sbin/nologin
saned:x:115:120::/var/lib/saned:/usr/sbin/nologin
colord:x:116:121:colord colour management daemon,,,:/var/lib/colord:/usr/sbin/nologin
pulse:x:117:122:PulseAudio daemon,,,:/run/pulse:/usr/sbin/nologin
cups-browsed:x:118:111::/nonexistent:/usr/sbin/nologin
xrdp:x:119:124::/run/xrdp:/usr/sbin/nologin
tom:x:1001:1004:tom,3,0322,2,1:/home/tom:/bin/bash
jerry:x:1005:1005:jerry,4,03222,03444,1:/home/jerry:/bin/bash
Scooby:x:1006:1006::/home/Scooby:/bin/sh
tooba65@ubuntu:~$

```

9. Change Scooby's shell to **/bin/bash** and verify.

Screenshot: task4_shell_switching.png

```

tooba65@ubuntu:~$ sudo usermod -s /bin/bash Scooby
tooba65@ubuntu:~$ su - Scooby
Password:
Scooby@ubuntu:~$ exit
logout
tooba65@ubuntu:~$

```

10. Create new groups **jolly** and **anime**.

Screenshot: task4_add_groups.png

```

tooba65@ubuntu:~$ sudo addgroup jolly
info: Selecting GID from range 1000 to 59999 ...
info: Adding group `jolly' (GID 1007) ...
tooba65@ubuntu:~$ sudo groupadd anime

```

11. Verify groups in **/etc/group**.

Screenshot: task4_verify_groups.png

```

tom:x:1001:tom
developer:x:1002:
devops:x:1003:
designer:x:1004:
jerry:x:1005:
Scooby:x:1006:
jolly:x:1007:
anime:x:1008:
tooba65@ubuntu:~$

```

12. Delete the created groups.

Screenshot: task4_delete_groups.png


```
tooba65@ubuntu:~$ sudo delgroup jolly
info: Removing group `jolly' ...
tooba65@ubuntu:~$ sudo groupdel anime
tooba65@ubuntu:~$ cat /etc/group
```

```
root:x:0:
docker:x:988:
tom:x:1001:tom
developer:x:1002:
devops:x:1003:
designer:x:1004:
jerry:x:1005:
Scooby:x:1006:
tooba65@ubuntu:~$
```

13. Delete the created users (Jerry and Scooby).

Screenshot: task4_delete_users.png

```
tooba65@ubuntu:~$ sudo deluser --remove-home Jerry
fatal: The user `Jerry' does not exist.
tooba65@ubuntu:~$ sudo deluser --remove-home jerry
info: Looking for files to backup/remove ...
info: Removing files ...
info: Removing crontab ...
info: Removing user `jerry' ...
tooba65@ubuntu:~$ sudo userdel -r Scooby
```

```
whoopsie:x:113:117::/nonexistent:/bin/false
avahi:x:114:118:Avahi mDNS daemon,,,:/run/avahi-daemon:/usr/sbin/nologin
saned:x:115:120::/var/lib/saned:/usr/sbin/nologin
colord:x:116:121:colord colour management daemon,,,:/var/lib/colord:/usr/sbin/nologin
pulse:x:117:122:PulseAudio daemon,,,:/run/pulse:/usr/sbin/nologin
cups-browsed:x:118:111::/nonexistent:/usr/sbin/nologin
xrdp:x:119:124::/run/xrdp:/usr/sbin/nologin
tom:x:1001:1004:tom,3,0322,2,1:/home/tom:/bin/bash
```

Task 5 – Create user Student; create files; set owner/group; identify file types

1. Create a user named **Student**.

Screenshot: task5_create_student.png


```

NAME_READ in configuration.
tooba65@ubuntu:~$ sudo adduser student
info: Adding user `student' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `student' (1005) ...
info: Adding new user `student' (1005) with group `student (1005)' ...
info: Creating home directory `/home/student' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for student
Enter the new value, or press ENTER for the default
    Full Name []: student
    Room Number []: 6
    Work Phone []: 0342
    Home Phone []: 0354
    Other []: 1
Is the information correct? [Y/n] y
info: Adding new user `student' to supplemental / extra groups `users' ...
info: Adding user `student' to group `users' ...

```

2. Switch to Student and create files and directories.

Screenshot: task5_create_files.png

```

tooba65@ubuntu:~$ su - student
Password:
student@ubuntu:~$ touch file1
student@ubuntu:~$ mkdir -p dir1
student@ubuntu:~$ touch dir1/file2
student@ubuntu:~$ ls -l
total 4
drwxrwxr-x 2 student student 4096 Nov  5 14:04 dir1
-rw-rw-r-- 1 student student   0 Nov  5 14:04 file1

```

3. Change the **owner** of file1 to tom.

Screenshot: task5_chown_file1.png

```

tooba65@ubuntu:~$ sudo chown tom /home/student/file1
tooba65@ubuntu:~$ ls -l /home/student/file1
ls: cannot access '/home/student/file1': Permission denied
tooba65@ubuntu:~$ ls -l file1
ls: cannot access 'file1': No such file or directory
tooba65@ubuntu:~$ sudo ls -l /home/student/file1
-rw-rw-r-- 1 tom student 0 Nov  5 14:04 /home/student/file1

```

4. Change the **group** of file1 to devops.

Screenshot: task5_chgrp_file1.png

```

tooba65@ubuntu:~$ sudo chgrp devops /home/student/file1
tooba65@ubuntu:~$ sudo ls -l /home/student/file1
-rw-rw-r-- 1 tom devops 0 Nov  5 14:04 /home/student/file1
tooba65@ubuntu:~$

```

5. Identify file types (file1, dir1, /dev/null).

Screenshot: task5_file_types.png

```

student@ubuntu:~$ ls -l
total 4
drwxrwxr-x 2 student student 4096 Nov  5 14:04 dir1
-rw-rw-r-- 1 tom      devops   0 Nov  5 14:04 file1
student@ubuntu:~$ ls -l dir1
total 0
-rw-rw-r-- 1 student student 0 Nov  5 14:04 file2
student@ubuntu:~$ ls -l /dev/null
crw-rw-rw- 1 root root 1, 3 Nov  5 13:09 /dev/null
student@ubuntu:~$ file file1 dir1 /dev/null
file1:      empty
dir1:       directory
/dev/null:  character special (1/3)
student@ubuntu:~$

```

6. Exit the Student user session.

Screenshot: task5_exit_student.png

```

student@ubuntu:~$ exit
logout
tooba65@ubuntu:~$

```

Task 6 – Change permissions using symbolic mode

1. Switch to Student and check file1.

Screenshot: task6_su_student.png

```

tooba65@ubuntu:~$ su - student
Password:
student@ubuntu:~$ cd ~
student@ubuntu:~$ ls -l file1
-rw-rw-r-- 1 tom devops 0 Nov  5 14:04 file1

```

2. Remove all permissions from file1.

Screenshot: task6_chmod_remove_rwx.png

```

tooba65@ubuntu:~$ sudo chown student:student /home/student/file1
tooba65@ubuntu:~$ sudo ls -l /home/student/file1
-rw-rw-r-- 1 student student 0 Nov  5 14:04 /home/student/file1
tooba65@ubuntu:~$ su - student
Password:
student@ubuntu:~$ cd ~
student@ubuntu:~$ chmod -rwx file1
student@ubuntu:~$ ls -l file1
----- 1 student student 0 Nov  5 14:04 file1
student@ubuntu:~$

```

3. Add read permission to all.

Screenshot: task6_chmod_add_r.png

```

student@ubuntu:~$ chmod +r file1
student@ubuntu:~$ ls -l file1
-r--r--r-- 1 student student 0 Nov  5 14:04 file1
student@ubuntu:~$

```

4. Add **execute** permission to the user.

Screenshot: task6_chmod_u_plus_x.png

```
student@ubuntu:~$ chmod u+x file1
student@ubuntu:~$ ls -l file1
-r-xr--r-- 1 student student 0 Nov  5 14:04 file1
student@ubuntu:~$
```

5. Add **write** permission to user and group.

Screenshot: task6_chmod_ug_plus_w.png

```
student@ubuntu:~$ chmod ug+w file1
student@ubuntu:~$ ls -l file1
-rwxrw-r-- 1 student student 0 Nov  5 14:04 file1
student@ubuntu:~$
```

6. Remove all permissions again explicitly.

Screenshot: task6_chmod_ugo_minus_rwx.png

```
student@ubuntu:~$ chmod ugo-rwx file1
student@ubuntu:~$ ls -l file1
----- 1 student student 0 Nov  5 14:04 file1
student@ubuntu:~$
```

Task 7 – Change permissions using “set” symbolic form (u= g= o=)

1. Switch to Student and verify file1.

Screenshot: task7_student_context.png

```
tooba65@ubuntu:~$ su - student
Password:
student@ubuntu:~$ cd ~
student@ubuntu:~$ ls -l file1
----- 1 student student 0 Nov  5 14:04 file1
student@ubuntu:~$
```

2. Set all permissions (u=g=o=rwx).

Screenshot: task7_chmod_set_all_rwx.png

```
student@ubuntu:~$ chmod u=rwx,g=rwx,o=rwx file1
student@ubuntu:~$ ls -l file1
-rwxrwxrwx 1 student student 0 Nov  5 14:04 file1
student@ubuntu:~$
```

3. Remove execute from group and others (g=rw, o=rw).

Screenshot: task7_remove_exec_go.png

```
student@ubuntu:~$ chmod g=rw,o=rw file1
student@ubuntu:~$ ls -l file1
-rwxrw-rw- 1 student student 0 Nov  5 14:04 file1
student@ubuntu:~$
```

4. Remove all permissions for all users.

Screenshot: task7_remove_all_perms.png

```
student@ubuntu:~$ chmod u=,g=,o= file1
student@ubuntu:~$ ls -l file1
----- 1 student student 0 Nov  5 14:04 file1
student@ubuntu:~$
```

Task 8 – Change permissions using numeric (octal) mode

1. Switch to Student and verify file1.

Screenshot: task8_student_context.png

```
tooba65@ubuntu:~$ su - student
Password:
student@ubuntu:~$ cd ~
student@ubuntu:~$ ls -l file1
----- 1 student student 0 Nov  5 14:04 file1
student@ubuntu:~$
```

2. Apply permission 777 and check.

Screenshot: task8_chmod_777.png

```
student@ubuntu:~$ chmod 777 file1
student@ubuntu:~$ ls -l file1
-rwxrwxrwx 1 student student 0 Nov  5 14:04 file1
student@ubuntu:~$
```

3. Apply permission 700 and check.

Screenshot: task8_chmod_700.png

```
student@ubuntu:~$ chmod 700 file1
student@ubuntu:~$ ls -l file1
-rwx----- 1 student student 0 Nov  5 14:04 file1
student@ubuntu:~$
```

4. Apply permission 744 and check.

Screenshot: task8_chmod_744.png

```
student@ubuntu:~$ chmod 744 file1
student@ubuntu:~$ ls -l file1
-rwxr--r-- 1 student student 0 Nov  5 14:04 file1
student@ubuntu:~$
```

5. Apply permission 640 and check.

Screenshot: task8_chmod_640.png

```
student@ubuntu:~$ chmod 640 file1
student@ubuntu:~$ ls -l file1
-rw-r----- 1 student student 0 Nov  5 14:04 file1
student@ubuntu:~$
```

6. Apply permission 664 and check.

Screenshot: task8_chmod_664.png

```
student@ubuntu:~$ chmod 664 file1
student@ubuntu:~$ ls -l file1
-rw-rw-r-- 1 student student 0 Nov  5 14:04 file1
student@ubuntu:~$
```

7. Apply permission 775 and check.

Screenshot: task8_chmod_775.png

```
student@ubuntu:~$ chmod 775 file1
student@ubuntu:~$ ls -l file1
-rwxrwxr-x 1 student student 0 Nov  5 14:04 file1
student@ubuntu:~$
```

8. Apply permission 750 and check.

Screenshot: task8_chmod_750.png

```
student@ubuntu:~$ chmod 750 file1
student@ubuntu:~$ ls -l file1
-rwxr-x--- 1 student student 0 Nov  5 14:04 file1
student@ubuntu:~$
```

Task 9 – Practice pipes, pagers, grep, and redirects with /var/log/syslog

1. View logs using less.

Screenshot: task9_grep_less.png

```
2025-11-02T11:00:36.404179+00:00 ubuntu systemd-modules-load[462]: Inserted module 'lp'
2025-11-02T11:00:36.418935+00:00 ubuntu systemd-modules-load[462]: Inserted module 'ppdev'
2025-11-02T11:00:36.419021+00:00 ubuntu systemd-modules-load[462]: Inserted module 'parport_pc'
2025-11-02T11:00:36.419033+00:00 ubuntu systemd-modules-load[462]: Inserted module 'msr'
2025-11-02T11:00:36.419040+00:00 ubuntu systemd-modules-load[462]: Inserted module 'dm_multipath'
2025-11-02T11:00:36.419047+00:00 ubuntu lvm[445]: 1 logical volume(s) in volume group "ubuntu-vg" monitored
2025-11-02T11:00:36.419054+00:00 ubuntu systemd[1]: Finished systemd-tmpfiles-setup-dev-early.service - Create Static Device Nodes in /dev gracefully.
2025-11-02T11:00:36.419063+00:00 ubuntu systemd[1]: Finished systemd-sysctl.service - Apply Kernel Variables.
2025-11-02T11:00:36.419163+00:00 ubuntu systemd[1]: Starting systemd-journal-flush.service - Flush Journal to Persistent Storage...
2025-11-02T11:00:36.419175+00:00 ubuntu systemd[1]: systemd-sysusers.service - Create System Users was skipped because no trigger condition checks were met.
2025-11-02T11:00:36.419182+00:00 ubuntu systemd[1]: Starting systemd-tmpfiles-setup-dev.service - Create Static Device Nodes in /dev...
2025-11-02T11:00:36.419189+00:00 ubuntu systemd[1]: Finished systemd-tmpfiles-setup-dev.service - Create Static Device Nodes in /dev.
2025-11-02T11:00:36.419197+00:00 ubuntu multipathd[480]: multipathd v0.9.4: start up
2025-11-02T11:00:36.419204+00:00 ubuntu multipathd[480]: reconfigure: setting up paths and maps
2025-11-02T11:00:36.419258+00:00 ubuntu multipathd[480]: sda: failed to get udev uid: No data available
2025-11-02T11:00:36.419266+00:00 ubuntu systemd[1]: Starting systemd-udevd.service - Rule-based Manager for Device Events and Files...
2025-11-02T11:00:36.419272+00:00 ubuntu systemd[1]: Started multipathd.service - Device-Mapper Multipath Device Controller.
2025-11-02T11:00:36.419278+00:00 ubuntu systemd[1]: Reached target local-fs-pre.target - Preparation for Local File Systems.
2025-11-02T11:00:36.419286+00:00 ubuntu systemd[1]: Mounting snap-code-211.mount - Mount unit for code, revision 211...
:
```

2. View logs using more.

Screenshot: task9_grep_more.png

```
tooba65@ubuntu:~$ cat /var/log/syslog | more
2025-11-02T11:00:36.404179+00:00 ubuntu systemd-modules-load[462]: Inserted module 'lp'
2025-11-02T11:00:36.418935+00:00 ubuntu systemd-modules-load[462]: Inserted module 'ppdev'
2025-11-02T11:00:36.419021+00:00 ubuntu systemd-modules-load[462]: Inserted module 'parport_pc'
2025-11-02T11:00:36.419033+00:00 ubuntu systemd-modules-load[462]: Inserted module 'msr'
2025-11-02T11:00:36.419040+00:00 ubuntu systemd-modules-load[462]: Inserted module 'dm_multipath'
2025-11-02T11:00:36.419047+00:00 ubuntu lvm[445]: 1 logical volume(s) in volume group "ubuntu-vg" monitored
2025-11-02T11:00:36.419054+00:00 ubuntu systemd[1]: Finished systemd-tmpfiles-setup-dev-early.service - Create Static Device Nodes in /dev gracefully.
2025-11-02T11:00:36.419063+00:00 ubuntu systemd[1]: Finished systemd-sysctl.service - Apply Kernel Variables.
2025-11-02T11:00:36.419163+00:00 ubuntu systemd[1]: Starting systemd-journal-flush.service - Flush Journal to Persistent Storage...
2025-11-02T11:00:36.419175+00:00 ubuntu systemd[1]: systemd-sysusers.service - Create System Users was skipped because no trigger condition checks were met.
2025-11-02T11:00:36.419182+00:00 ubuntu systemd[1]: Starting systemd-tmpfiles-setup-dev.service - Create Static Device Nodes in /dev...
2025-11-02T11:00:36.419189+00:00 ubuntu systemd[1]: Finished systemd-tmpfiles-setup-dev.service - Create Static Device Nodes in /dev.
2025-11-02T11:00:36.419197+00:00 ubuntu multipathd[480]: multipathd v0.9.4: start up
2025-11-02T11:00:36.419204+00:00 ubuntu multipathd[480]: reconfigure: setting up paths and maps
2025-11-02T11:00:36.419258+00:00 ubuntu multipathd[480]: sda: failed to get udev uid: No data available
2025-11-02T11:00:36.419266+00:00 ubuntu systemd[1]: Starting systemd-udevd.service - Rule-based Manager for Device Events and Files...
2025-11-02T11:00:36.419272+00:00 ubuntu systemd[1]: Started multipathd.service - Device-Mapper Multipath Device Controller.
2025-11-02T11:00:36.419278+00:00 ubuntu systemd[1]: Reached target local-fs-pre.target - Preparation for Local File Systems.
2025-11-02T11:00:36.419286+00:00 ubuntu systemd[1]: Mounting snap-code-211.mount - Mount unit for code, revision 211...
--More--
```

3. Use grep to search for errors or failures.

Screenshot: task9_grep_head.png

```
tooba65@ubuntu:~$ sudo grep -E 'fail|error' /var/log/syslog | head
2025-11-02T11:00:36.419258+00:00 ubuntu multipathd[480]: sda: failed to get udev uid: No data available
2025-11-02T11:00:36.419540+00:00 ubuntu multipathd[480]: sda: failed to get sysfs uid: No such file or directory
2025-11-02T11:00:36.419548+00:00 ubuntu multipathd[480]: sda: failed to get sgio uid: No such file or directory
2025-11-02T11:00:36.419616+00:00 ubuntu multipathd[480]: sda: failed to get udev uid: No data available
2025-11-02T11:00:36.419623+00:00 ubuntu multipathd[480]: sda: failed to get path uid
2025-11-02T11:00:36.419630+00:00 ubuntu multipathd[480]: uevent trigger error
2025-11-02T11:00:36.420999+00:00 ubuntu systemd[1]: apport-autoreport.path - Process error reports when automatic reporting is enabled (file watch) was skipped because of an unmet condition check (ConditionPathExists=/var/lib/apport/autoreport).
2025-11-02T11:00:36.421066+00:00 ubuntu systemd[1]: apport-autoreport.timer - Process error reports when automatic reporting is enabled (timer based) was skipped because of an unmet condition check (ConditionPathExists=/var/lib/apport/autoreport).
2025-11-02T11:00:36.423032+00:00 ubuntu systemd[1]: Starting grub-initrd-fallback.service - GRUB failed boot detection...
2025-11-02T11:00:36.423410+00:00 ubuntu systemd[1]: Finished grub-initrd-fallback.service - GRUB failed boot detection.
tooba65@ubuntu:~$
```

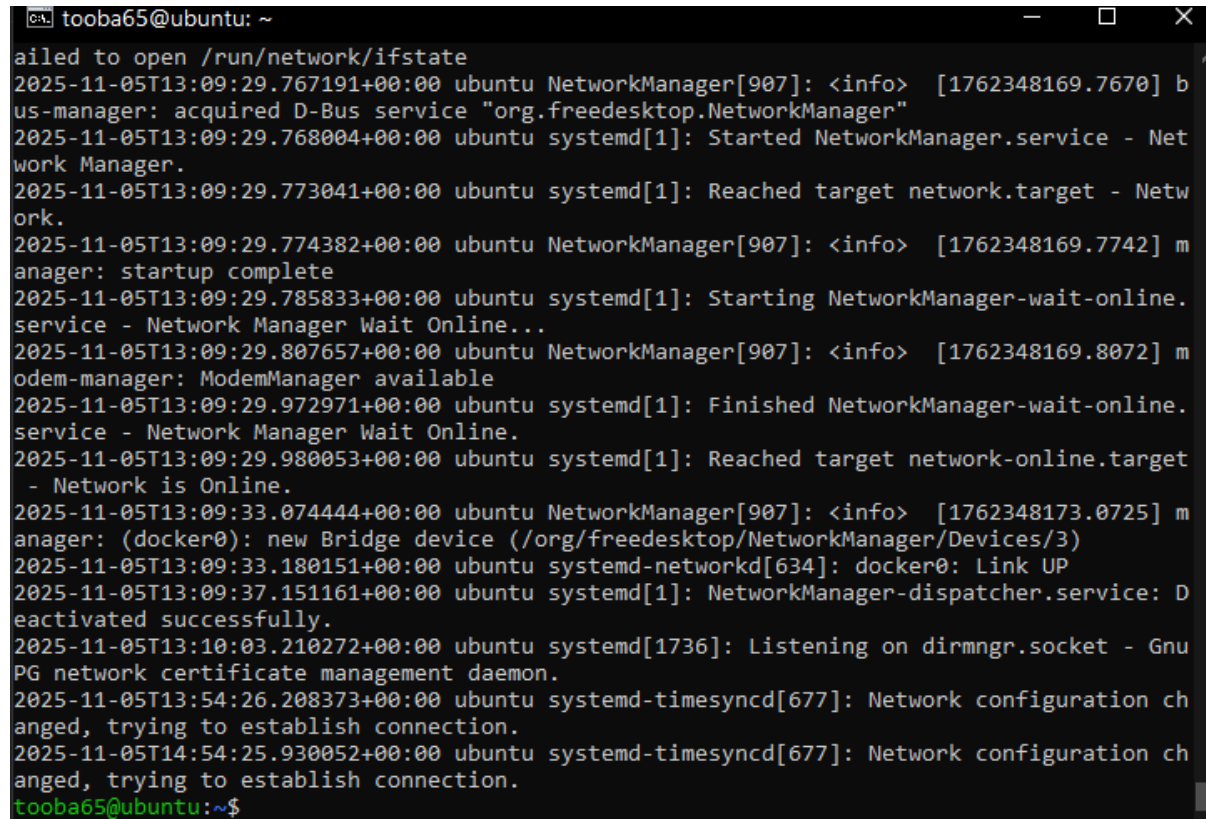

4. Redirect systemd logs to a file (overwrite).

Screenshot: task9_redirect_overwrite.png

```
tooba65@ubuntu:~$ sudo grep -i systemd /var/log/syslog > ~/syslog_systemd.txt
```

5. Append network logs to the same file and view.

Screenshot: task9_redirect_append.png



```
tooba65@ubuntu: ~
ailed to open /run/network/ifstate
2025-11-05T13:09:29.767191+00:00 ubuntu NetworkManager[907]: <info> [1762348169.7670] b
us-manager: acquired D-Bus service "org.freedesktop.NetworkManager"
2025-11-05T13:09:29.768004+00:00 ubuntu systemd[1]: Started NetworkManager.service - Net
work Manager.
2025-11-05T13:09:29.773041+00:00 ubuntu systemd[1]: Reached target network.target - Netw
ork.
2025-11-05T13:09:29.774382+00:00 ubuntu NetworkManager[907]: <info> [1762348169.7742] m
anager: startup complete
2025-11-05T13:09:29.785833+00:00 ubuntu systemd[1]: Starting NetworkManager-wait-online.
service - Network Manager Wait Online...
2025-11-05T13:09:29.807657+00:00 ubuntu NetworkManager[907]: <info> [1762348169.8072] m
odem-manager: ModemManager available
2025-11-05T13:09:29.972971+00:00 ubuntu systemd[1]: Finished NetworkManager-wait-online.
service - Network Manager Wait Online.
2025-11-05T13:09:29.980053+00:00 ubuntu systemd[1]: Reached target network-online.target
- Network is Online.
2025-11-05T13:09:33.074444+00:00 ubuntu NetworkManager[907]: <info> [1762348173.0725] m
anager: (docker0): new Bridge device (/org/freedesktop/NetworkManager/Devices/3)
2025-11-05T13:09:33.180151+00:00 ubuntu systemd-networkd[634]: docker0: Link UP
2025-11-05T13:09:37.151161+00:00 ubuntu systemd[1]: NetworkManager-dispatcher.service: D
eactivated successfully.
2025-11-05T13:10:03.210272+00:00 ubuntu systemd[1736]: Listening on dirmngr.socket - Gnu
PG network certificate management daemon.
2025-11-05T13:54:26.208373+00:00 ubuntu systemd-timesyncd[677]: Network configuration ch
anged, trying to establish connection.
2025-11-05T14:54:25.930052+00:00 ubuntu systemd-timesyncd[677]: Network configuration ch
anged, trying to establish connection.
tooba65@ubuntu:~$
```

Task 10 – Script setup.sh – variables, command substitution, file/dir checks, permissions

Step 1 – Add shebang

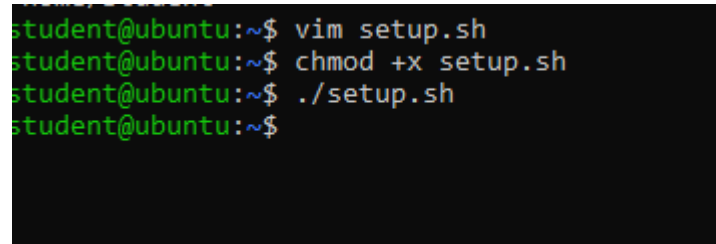
- Open vim and add:
 - `#!/bin/bash`
- Save and quit (`:wq`)
- Make it executable: `chmod +x setup.sh`
- Run script: `./setup.sh`

Screenshots

- task10_b1_vim.png

```
#!/bin/bash
```


- task10_b1_run.png



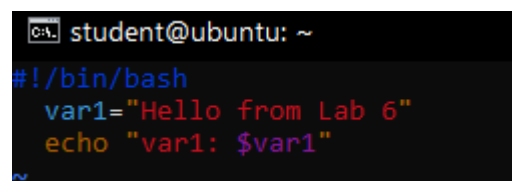
```
student@ubuntu:~$ vim setup.sh
student@ubuntu:~$ chmod +x setup.sh
student@ubuntu:~$ ./setup.sh
student@ubuntu:~$
```

Step 2 – Define variable var1 and echo it

- Open setup.sh in vim.
- Append code for defining var1 and echo it.
- Save and quit.
- Run: ./setup.sh

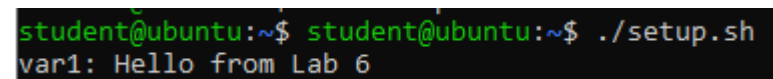
Screenshots

- task10_b2_vim.png



```
student@ubuntu: ~
#!/bin/bash
var1="Hello from Lab 6"
echo "var1: $var1"
```

- task10_b2_run.png



```
student@ubuntu:~$ student@ubuntu:~$ ./setup.sh
var1: Hello from Lab 6
```

Step 3 – Save output of ls -l into variable allFiles and echo it

- Open vim.
- Append the code for allFiles="\$(ls -l)".
- Save and quit.
- Run: ./setup.sh

Screenshots

- task10_b3_vim.png

```
#!/bin/bash
var1="Hello from Lab 6"
echo "var1: $var1"
allFiles="$(ls -l)"
echo "allFiles (ls -l):"
echo "$allFiles"
```

- task10_b3_run.png

```
student@ubuntu:~$ student@ubuntu:~$ ./setup.sh
var1: Hello from Lab 6
allFiles (ls -l):
total 8
drwxrwxr-x 2 student student 4096 Nov  5 14:04 dir1
-rwxr-x--- 1 student student   0 Nov  5 14:04 file1
-rwxrwxr-x 1 student student  127 Nov  5 15:29 setup.sh
student@ubuntu:~$
```

Step 4 – If directory dir1 exists else create it

- Open vim.
- Append the directory-check code.
- Save and quit.
- Run: ./setup.sh

Screenshots

- task10_b4_vim.png

```
student@ubuntu: ~
#!/bin/bash
var1="Hello from Lab 6"
echo "var1: $var1"
allFiles="$(ls -l)"
echo "allFiles (ls -l):"
echo "$allFiles"
# Directory check
if [ -d "dir1" ]; then
    echo "Directory dir1 exists."
else
    echo "Directory dir1 does not exist. Creating..."
    mkdir -p "dir1"
    echo "Directory dir1 created."
fi
```

- task10_b4_run.png

```

student@ubuntu:~$ student@ubuntu:~$ ./setup.sh
var1: Hello from Lab 6
allFiles (ls -l):
total 8
drwxrwxr-x 2 student student 4096 Nov  5 14:04 dir1
-rwxr-x--- 1 student student   0 Nov  5 14:04 file1
-rwxrwxr-x 1 student student  340 Nov  5 15:32 setup.sh
Directory dir1 exists.
student@ubuntu:~$

```

Step 5 – If file dir1/file2 does not exist create it

- Open vim.
- Append the file-check code for dir1/file2.
- Save and quit.
- Run: ./setup.sh

Screenshots

- task10_b5_vim.png

```

# File check
if [ -f "dir1/file2" ]; then
    echo "file2 already exists."
else
    echo "file2 does not exist. Creating..."
    touch "dir1/file2"
    chmod a-rwx "dir1/file2"
    echo "file2 created."
fi

```

- task10_b5_run.png

```

student@ubuntu:~$ student@ubuntu:~$ ./setup.sh
var1: Hello from Lab 6
allFiles (ls -l):
total 8
drwxrwxr-x 2 student student 4096 Nov  5 14:04 dir1
-rwxr-x--- 1 student student   0 Nov  5 14:04 file1
-rwxrwxr-x 1 student student  566 Nov  5 15:34 setup.sh
Directory dir1 exists.
file2 already exists.
student@ubuntu:~$

```

Step 6 – Check permissions on dir1/file2 and grant missing user permissions

- Open vim.
- Append the permission-check code.
- Save and quit.
- Run: ./setup.sh

Screenshots

- task10_b6_vim.png

```
# Permission checks for dir1/file2 (user permissions)
f="dir1/file2"
if [ ! -r "$f" ]; then
    echo "Read permission missing; granting to user..."
    chmod u+r "$f"
fi

if [ ! -w "$f" ]; then
    echo "Write permission missing; granting to user..."
    chmod u+w "$f"
fi

if [ ! -x "$f" ]; then
    echo "Execute permission missing; granting to user..."
    chmod u+x "$f"
fi

echo "Final permissions for $f:"
ls -l "$f"

-- INSERT --
```

- task10_b6_run.png

```
student@ubuntu:~$ student@ubuntu:~$ ./setup.sh
var1: Hello from Lab 6
allFiles (ls -l):
total 8
drwxrwxr-x 2 student student 4096 Nov  5 14:04 dir1
-rwxr-x--- 1 student student   0 Nov  5 14:04 file1
-rwxrwxr-x 1 student student  997 Nov  5 15:37 setup.sh
Directory dir1 exists.
file2 already exists.
Execute permission missing; granting to user...
Final permissions for dir1/file2:
-rwxrw-r-- 1 student student 0 Nov  5 14:04 dir1/file2
student@ubuntu:~$
```

Task 11 – Script setup.sh – argument comparisons and string checks

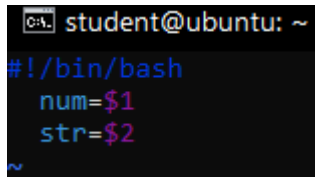
Step 0 – Create file with shebang and set num and str variables

- Open vim, overwrite setup.sh.
- Insert shebang and num=\$1, str=\$2.
- Save and quit.
- Make executable and run:
 - chmod +x setup.sh

- ./setup.sh 10 Student

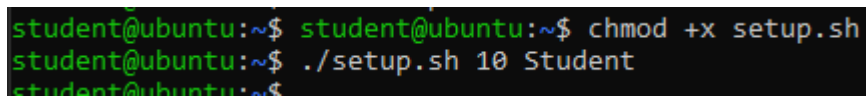
Screenshots

- task11_b0_vim.png



```
student@ubuntu: ~  
#!/bin/bash  
num=$1  
str=$2  
~
```

- task11_b0_run.png



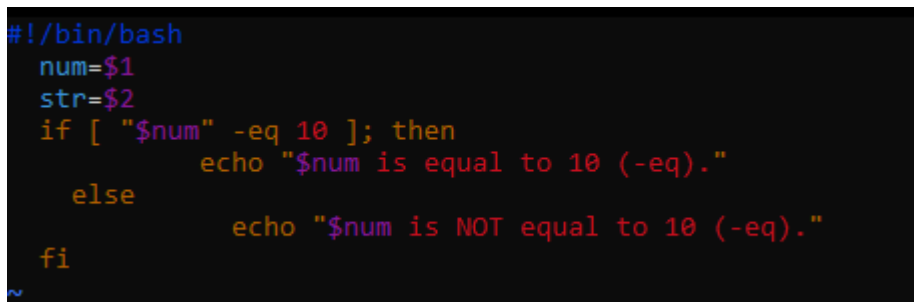
```
student@ubuntu:~$ student@ubuntu:~$ chmod +x setup.sh  
student@ubuntu:~$ ./setup.sh 10 Student  
student@ubuntu:~$
```

Step 1 – Add -eq (equal) test

- Open vim, append -eq if-statement.
- Save and quit.
- Run:
 - ./setup.sh 10 Student
 - ./setup.sh 7 Student

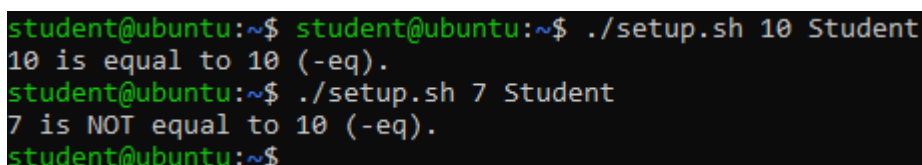
Screenshots

- task11_b1_vim.png



```
#!/bin/bash  
num=$1  
str=$2  
if [ "$num" -eq 10 ]; then  
    echo "$num is equal to 10 (-eq)."  
else  
    echo "$num is NOT equal to 10 (-eq)."  
fi  
~
```

- task11_b1_run.png



```
student@ubuntu:~$ student@ubuntu:~$ ./setup.sh 10 Student  
10 is equal to 10 (-eq).  
student@ubuntu:~$ ./setup.sh 7 Student  
7 is NOT equal to 10 (-eq).  
student@ubuntu:~$
```

Step 2 – Add -ne (not equal) test

- Append -ne if-statement.

- Run:
 - ./setup.sh 7 Student
 - ./setup.sh 10 Student

Screenshots

- task11_b2_vim.png

```
#!/bin/bash
num=$1
str=$2
if [ "$num" -eq 10 ]; then
    echo "$num is equal to 10 (-eq)."
else
    echo "$num is NOT equal to 10 (-eq)."
fi
if [ "$num" -ne 10 ]; then
    echo "$num is not equal to 10 (-ne)."
else
    echo "$num is equal to 10 (-ne false)."
fi
```

- task11_b2_run.png

```
student@ubuntu:~$ student@ubuntu:~$ ./setup.sh 7 Student
7 is NOT equal to 10 (-eq).
7 is not equal to 10 (-ne).
student@ubuntu:~$ ./setup.sh 10 Student
10 is equal to 10 (-eq).
10 is equal to 10 (-ne false).
student@ubuntu:~$
```

Step 3 – Add -gt (greater than) test

- Append -gt if-statement.
- Run:
 - ./setup.sh 12 Student
 - ./setup.sh 9 Student

Screenshots

- task11_b3_vim.png


```
student@ubuntu: ~  
#!/bin/bash  
num=$1  
str=$2  
if [ "$num" -eq 10 ]; then  
    echo "$num is equal to 10 (-eq)."  
else  
    echo "$num is NOT equal to 10 (-eq)."  
fi  
if [ "$num" -ne 10 ]; then  
    echo "$num is not equal to 10 (-ne)."  
else  
    echo "$num is equal to 10 (-ne false)."  
fi  
if [ "$num" -gt 10 ]; then  
    echo "$num is greater than 10 (-gt)."  
else  
    echo "$num is NOT greater than 10 (-gt)."  
fi
```

- task11_b3_run.png

```
student@ubuntu:~$ ./setup.sh 12 Student  
12 is NOT equal to 10 (-eq).  
12 is not equal to 10 (-ne).  
12 is greater than 10 (-gt).  
student@ubuntu:~$ ./setup.sh 9 Student  
9 is NOT equal to 10 (-eq).  
9 is not equal to 10 (-ne).  
9 is NOT greater than 10 (-gt).  
student@ubuntu:~$
```

Step 4 – Add -lt (less than) test

- Append -lt if-statement.
- Run:
 - ./setup.sh 5 Student
 - ./setup.sh 11 Student

Screenshots

- task11_b4_vim.png

```
student@ubuntu: ~  
num=$1  
str=$2  
if [ "$num" -eq 10 ]; then  
    echo "$num is equal to 10 (-eq)."  
else  
    echo "$num is NOT equal to 10 (-eq)."  
fi  
if [ "$num" -ne 10 ]; then  
    echo "$num is not equal to 10 (-ne)."  
else  
    echo "$num is equal to 10 (-ne false)."  
fi  
if [ "$num" -gt 10 ]; then  
    echo "$num is greater than 10 (-gt)."  
else  
    echo "$num is NOT greater than 10 (-gt)."  
fi  
if [ "$num" -lt 10 ]; then  
    echo "$num is less than 10 (-lt)."  
else  
    echo "$num is NOT less than 10 (-lt)."  
fi
```

- task11_b4_run.png

```
student@ubuntu:~$ student@ubuntu:~$ ./setup.sh 5 Student  
5 is NOT equal to 10 (-eq).  
5 is not equal to 10 (-ne).  
5 is NOT greater than 10 (-gt).  
5 is less than 10 (-lt).  
student@ubuntu:~$ ./setup.sh 11 Student  
11 is NOT equal to 10 (-eq).  
11 is not equal to 10 (-ne).  
11 is greater than 10 (-gt).  
11 is NOT less than 10 (-lt).
```

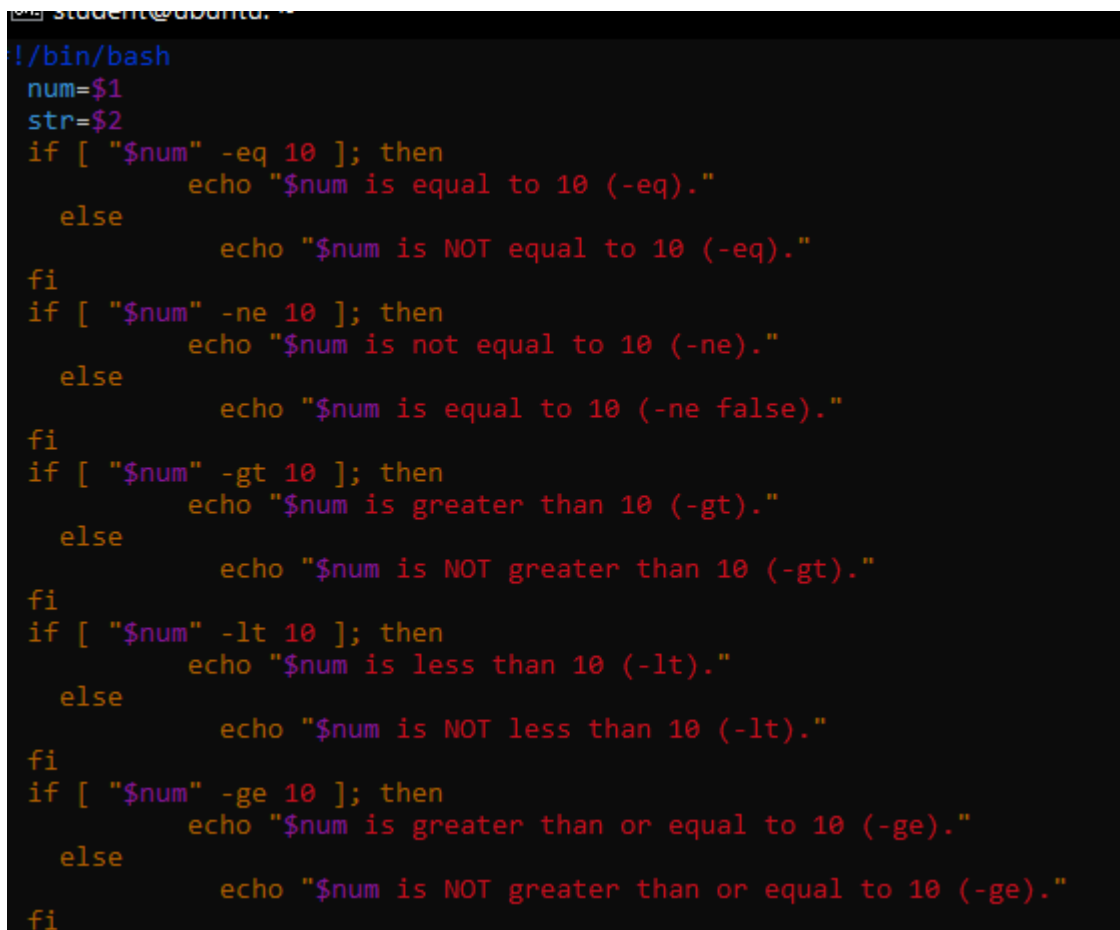
Step 5 – Add -ge (greater or equal) test

- Append -ge if-statement.
- Run:
 - ./setup.sh 10 Student
 - ./setup.sh 8 Student

Screenshots

- task11_b5_vim.png

```
student@ubuntu:~$ cat setup.sh
#!/bin/bash
num=$1
str=$2
if [ "$num" -eq 10 ]; then
    echo "$num is equal to 10 (-eq)."
```



```
    else
        echo "$num is NOT equal to 10 (-eq)."
```

```
fi
if [ "$num" -ne 10 ]; then
    echo "$num is not equal to 10 (-ne)."
```

```
    else
        echo "$num is equal to 10 (-ne false)."
```

```
fi
if [ "$num" -gt 10 ]; then
    echo "$num is greater than 10 (-gt)."
```

```
    else
        echo "$num is NOT greater than 10 (-gt)."
```

```
fi
if [ "$num" -lt 10 ]; then
    echo "$num is less than 10 (-lt)."
```

```
    else
        echo "$num is NOT less than 10 (-lt)."
```

```
fi
if [ "$num" -ge 10 ]; then
    echo "$num is greater than or equal to 10 (-ge)."
```

```
    else
        echo "$num is NOT greater than or equal to 10 (-ge)."
```

```
fi
```

- task11_b5_run.png

```
student@ubuntu:~$ ./setup.sh 10 Student
10 is equal to 10 (-eq).
10 is equal to 10 (-ne false).
10 is NOT greater than 10 (-gt).
10 is NOT less than 10 (-lt).
10 is greater than or equal to 10 (-ge).
student@ubuntu:~$ ./setup.sh 8 Student
8 is NOT equal to 10 (-eq).
8 is not equal to 10 (-ne).
8 is NOT greater than 10 (-gt).
8 is less than 10 (-lt).
8 is NOT greater than or equal to 10 (-ge).
```

Step 6 – Add -le (less or equal) test

- Append -le if-statement.
- Run:
 - ./setup.sh 10 Student
 - ./setup.sh 12 Student

Screenshots

- task11_b6_vim.png

```

fi
if [ "$num" -le 10 ]; then
    echo "$num is less than or equal to 10 (-le)."

```

- task11_b6_run.png

```

student@ubuntu:~$ ./setup.sh 10 Student
10 is equal to 10 (-eq).
10 is equal to 10 (-ne false).
10 is NOT greater than 10 (-gt).
10 is NOT less than 10 (-lt).
10 is greater than or equal to 10 (-ge).
10 is less than or equal to 10 (-le).
student@ubuntu:~$ ./setup.sh 12 Student
12 is NOT equal to 10 (-eq).
12 is not equal to 10 (-ne).
12 is greater than 10 (-gt).
12 is NOT less than 10 (-lt).
12 is greater than or equal to 10 (-ge).
12 is NOT less than or equal to 10 (-le).
student@ubuntu:~$
```

Step 7 – String equality (=)

- Append string equality test using ["\$str" = "Student"].
- Run:
 - ./setup.sh 10 Student
 - ./setup.sh 10 Test

Screenshots

- task11_b7_vim.png

```

if [ "$str" = "Student" ]; then
    echo "Second argument equals 'Student' ( = )."

```

- task11_b7_run.png

```

student@ubuntu:~$ student@ubuntu:~$ ./setup.sh 10 Student
10 is equal to 10 (-eq).
10 is equal to 10 (-ne false).
10 is NOT greater than 10 (-gt).
10 is NOT less than 10 (-lt).
10 is greater than or equal to 10 (-ge).
10 is less than or equal to 10 (-le).
Second argument equals 'Student' ( = ).
student@ubuntu:~$ ./setup.sh 10 Test
10 is equal to 10 (-eq).
10 is equal to 10 (-ne false).
10 is NOT greater than 10 (-gt).
10 is NOT less than 10 (-lt).
10 is greater than or equal to 10 (-ge).
10 is less than or equal to 10 (-le).
Second argument does NOT equal 'Student' ( = ).
student@ubuntu:~$

```

Step 8 – String inequality (!=)

- Append ["\$str" != "Student"] test.
- Run:
 - ./setup.sh 10 Test
 - ./setup.sh 10 Student

Screenshots

- task11_b8_vim.png

```

fi
if [ "$str" != "Student" ]; then
    echo "Second argument is not equal to 'Student' ( != )."
else
    echo "Second argument equals 'Student' ( != false)."
fi
- INSERT --
43

```

- task11_b8_run.png

```
student@ubuntu:~$ student@ubuntu:~$ ./setup.sh 10 Test
10 is equal to 10 (-eq).
10 is equal to 10 (-ne false).
10 is NOT greater than 10 (-gt).
10 is NOT less than 10 (-lt).
10 is greater than or equal to 10 (-ge).
10 is less than or equal to 10 (-le).
Second argument does NOT equal 'Student' ( = ).
Second argument is not equal to 'Student' ( != ).
student@ubuntu:~$ ./setup.sh 10 Student
10 is equal to 10 (-eq).
10 is equal to 10 (-ne false).
10 is NOT greater than 10 (-gt).
10 is NOT less than 10 (-lt).
10 is greater than or equal to 10 (-ge).
10 is less than or equal to 10 (-le).
Second argument equals 'Student' ( = ).
Second argument equals 'Student' ( != false).
student@ubuntu:~$
```

Step 9 – Check if second argument is empty (-z)

- Append [-z "\$str"] test.
- Run:
 - ./setup.sh 10
 - ./setup.sh 10 Student

Screenshots

- task11_b9_vim.png

```
fi
if [ -z "$str" ]; then
    echo "Second argument is empty (zero-length)."
else
    echo "Second argument is not empty."
fi
```

- task11_b9_run.png

```

student@ubuntu:~$ student@ubuntu:~$ ./setup.sh 10
10 is equal to 10 (-eq).
10 is equal to 10 (-ne false).
10 is NOT greater than 10 (-gt).
10 is NOT less than 10 (-lt).
10 is greater than or equal to 10 (-ge).
10 is less than or equal to 10 (-le).
Second argument does NOT equal 'Student' ( = ).
Second argument is not equal to 'Student' ( != ).
Second argument is empty (zero-length).
student@ubuntu:~$ ./setup.sh 10 Student
10 is equal to 10 (-eq).
10 is equal to 10 (-ne false).
10 is NOT greater than 10 (-gt).
10 is NOT less than 10 (-lt).
10 is greater than or equal to 10 (-ge).
10 is less than or equal to 10 (-le).
Second argument equals 'Student' ( = ).
Second argument equals 'Student' ( != false).
Second argument is not empty.
student@ubuntu:~$

```

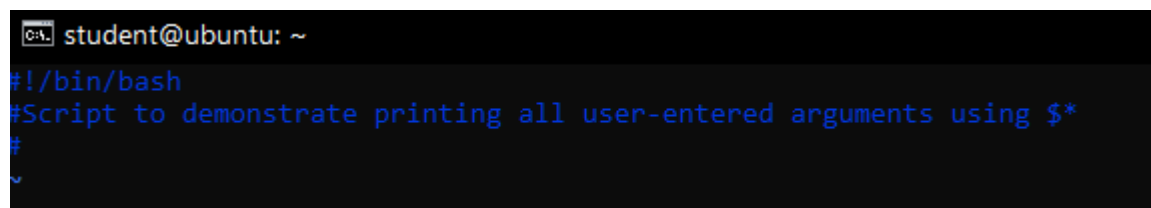
Task 12 – Script setup.sh – print all arguments with a for loop

Step 1 – Create file with shebang and comment

- Open vim and add shebang and a short comment.
- Save and quit.
- Run ./setup.sh

Screenshots

- task12_b1_vim.png

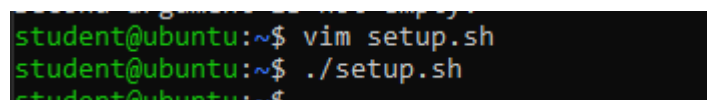


```

C:\ student@ubuntu: ~
#!/bin/bash
#Script to demonstrate printing all user-entered arguments using $*
#
~
~

```

- task12_b1_run.png



```

student@ubuntu:~$ vim setup.sh
student@ubuntu:~$ ./setup.sh
student@ubuntu:~$

```

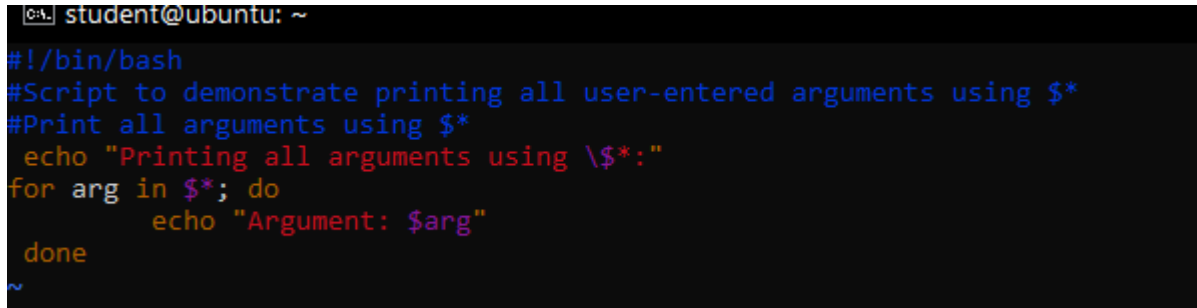
Step 2 – Append for-loop using \$*

- Open vim, append the loop that echoes each argument.
- Save and quit.

- Run:
 - `chmod +x setup.sh`
 - `./setup.sh one "two words" three`

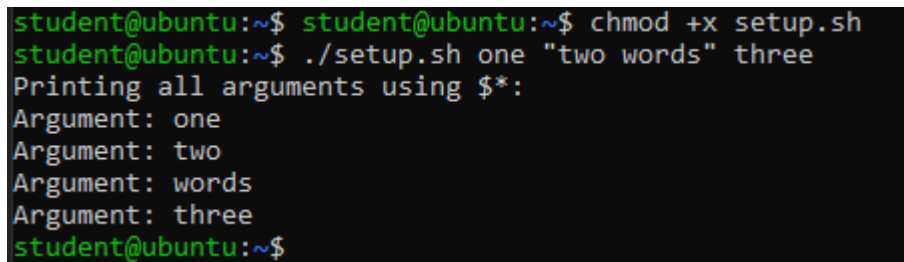
Screenshots

- task12_b2_vim.png



```
student@ubuntu: ~  
#!/bin/bash  
#Script to demonstrate printing all user-entered arguments using $*  
#Print all arguments using $*  
echo "Printing all arguments using \${$*}:"  
for arg in $*; do  
    echo "Argument: $arg"  
done  
~
```

- task12_b2_run.png



```
student@ubuntu:~$ student@ubuntu:~$ chmod +x setup.sh  
student@ubuntu:~$ ./setup.sh one "two words" three  
Printing all arguments using $*:  
Argument: one  
Argument: two  
Argument: words  
Argument: three  
student@ubuntu:~$
```

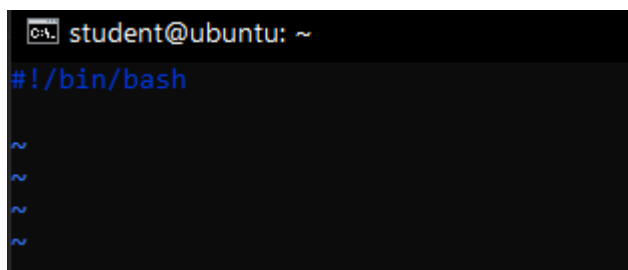
Task 13 – Script setup.sh – while loop summation and functions

Step 1 – Add shebang

- Overwrite setup.sh and add shebang.
- Save and quit.
- Run:
 - `chmod +x setup.sh`
 - `./setup.sh`

Screenshots

- task13_b1_vim.png



```
student@ubuntu: ~  
#!/bin/bash  
~  
~  
~  
~  
~
```

- task13_b1_run.png

```
student@ubuntu:~$ student@ubuntu:~$ chmod +x setup.sh
student@ubuntu:~$ ./setup.sh
student@ubuntu:~$
```

Step 2 – Add while-loop summation

- Open vim, append interactive while-loop for numbers.
- Save and quit.
- Run ./setup.sh and enter:
 - 5
 - 7
 - q

Screenshots

- task13_b2_vim.png

```
student@ubuntu: ~
#!/bin/bash
sum=0
while true; do
    read -p "Enter a number (or 'q' to quit): " input
    if [ "$input" = "q" ]; then
        break
    fi

    sum=$((sum + input))
    echo "Total Score: $sum"
done
echo "Final total: $sum"
~
```

- task13_b2_run.png

```
student@ubuntu:~$ student@ubuntu:~$ ./setup.sh
Enter a number (or 'q' to quit): 5
Total Score: 5
Enter a number (or 'q' to quit): 7
Total Score: 12
Enter a number (or 'q' to quit): q
Final total: 12
student@ubuntu:~$
```

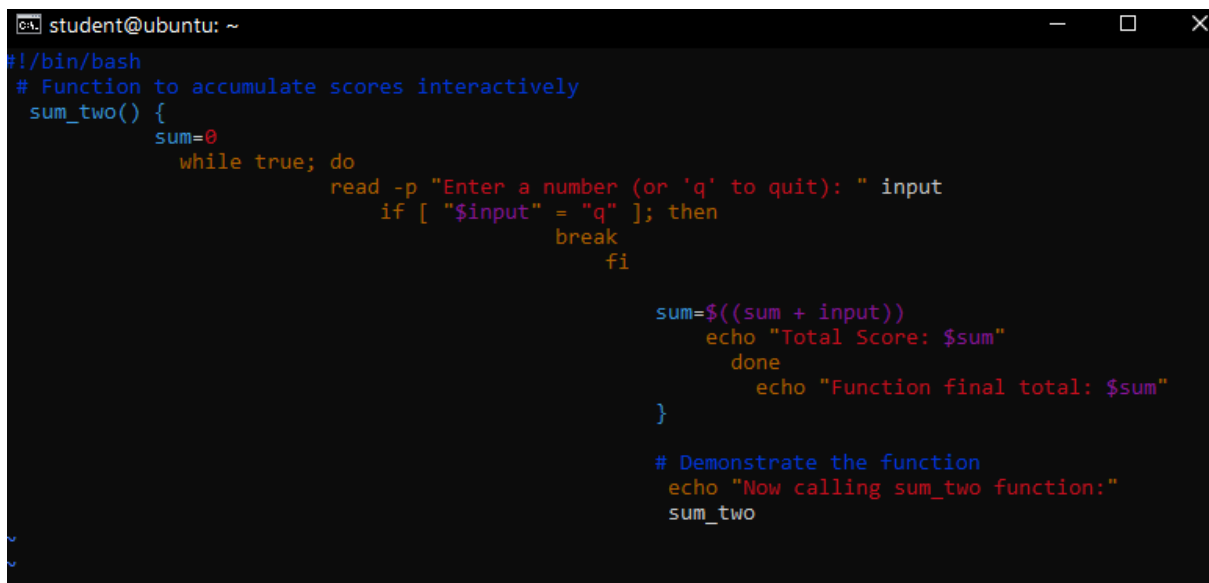
Step 3 – Add interactive function sum_two()

- Open vim, append function with interactive loop.

- Save and quit.
- Run ./setup.sh and enter:
 - 3
 - 4
 - q

Screenshots

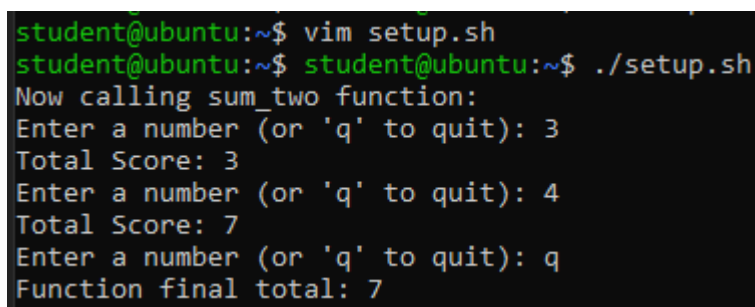
- task13_b3_vim.png



```
student@ubuntu: ~
#!/bin/bash
# Function to accumulate scores interactively
sum_two() {
    sum=0
    while true; do
        read -p "Enter a number (or 'q' to quit): " input
        if [ "$input" = "q" ]; then
            break
        fi
        sum=$((sum + input))
        echo "Total Score: $sum"
    done
    echo "Function final total: $sum"
}

# Demonstrate the function
echo "Now calling sum_two function:"
sum_two
```

- task13_b3_run.png



```
student@ubuntu:~$ vim setup.sh
student@ubuntu:~$ student@ubuntu:~$ ./setup.sh
Now calling sum_two function:
Enter a number (or 'q' to quit): 3
Total Score: 3
Enter a number (or 'q' to quit): 4
Total Score: 7
Enter a number (or 'q' to quit): q
Function final total: 7
```

Step 4 – Add function that takes two arguments and returns sum

- Open vim, append the sum_args function and demo call.
- Save and quit.
- Run ./setup.sh

Screenshots

- task13_b4_vim.png

```
# Function that sums two arguments and returns the result
sum_args() {
    a=$1
    b=$2
    return $((a + b))
}
```

```
Demonstrate sum_args function
echo "Now demonstrating sum_args function:"
sum_args 3 4
result=$?
echo "sum_args(3,4) returned: $result"
```

```
# Function to accumulate scores interactively
um_two() {
    sum=0
    while true; do
        read -p "Enter a number (or 'q' to quit): " input
        if [ "$input" = "q" ]; then
            break
        fi
        sum=$((sum + input))
        echo "Total Score: $sum"
    done
    echo "Function final total: $sum"
}

# Demonstrate the interactive summation function
echo "Now calling sum_two function:"
sum_two
# Function that sums two arguments and returns the result
sum_args() {
    a=$1
    b=$2
    return $((a + b))
}

# Demonstrate sum_args function
echo "Now demonstrating sum_args function:"
sum_args 3 4
result=$?
echo "sum_args(3,4) returned: $result"
```

INSERT --

- task13_b4_run.png

```
student@ubuntu:~$ vim setup.sh
student@ubuntu:~$ student@ubuntu:~$ chmod +x setup.sh
student@ubuntu:~$ ./setup.sh
Now calling sum_two function:
Enter a number (or 'q' to quit): 1
Total Score: 1
Enter a number (or 'q' to quit): 2
Total Score: 3
Enter a number (or 'q' to quit): q
Function final total: 3
Now demonstrating sum_args function:
sum_args(3,4) returned: 7
student@ubuntu:~$
```

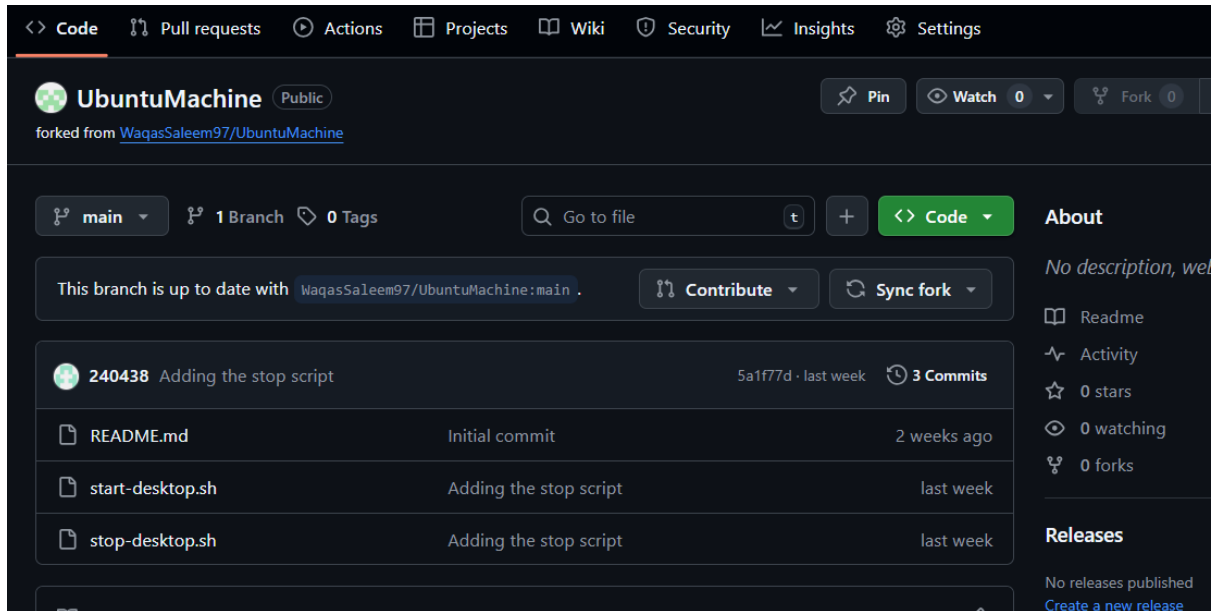
Task 14 – Codespaces GUI — fork repo, run start-desktop.sh, open VNC, stop GUI

Step 1 – Fork repository

- Open the repo URL.
- Click **Fork** to your account.

Screenshot

- task14_fork.png

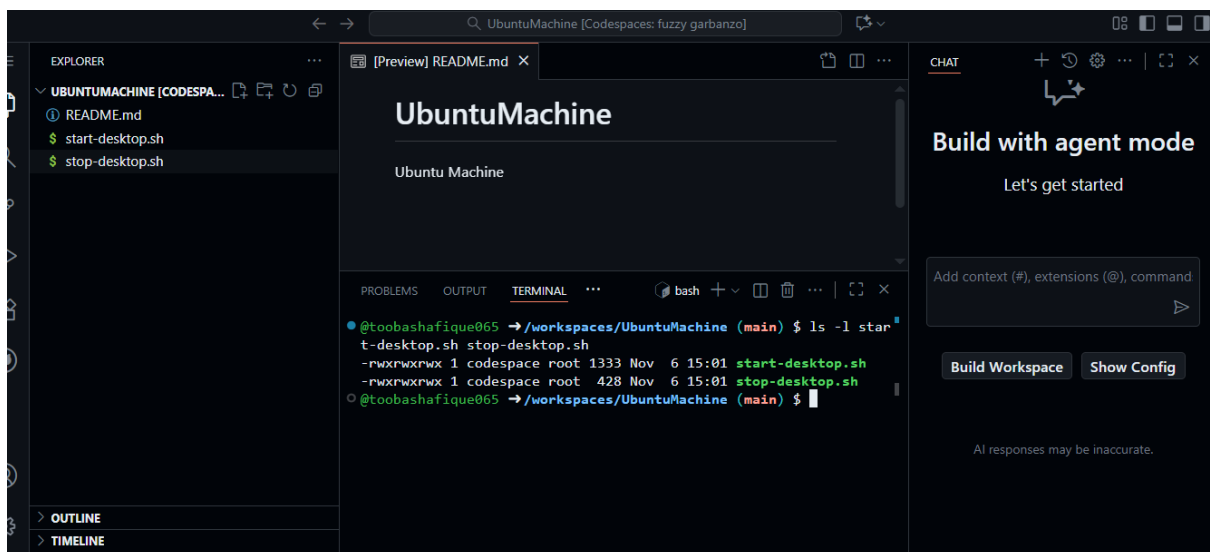


Step 2 – Open Codespace

- On your forked repo, click **Code** > **Open with Codespaces** > **Create codespace on main**.

Screenshot

- task14_codespace_launch.png

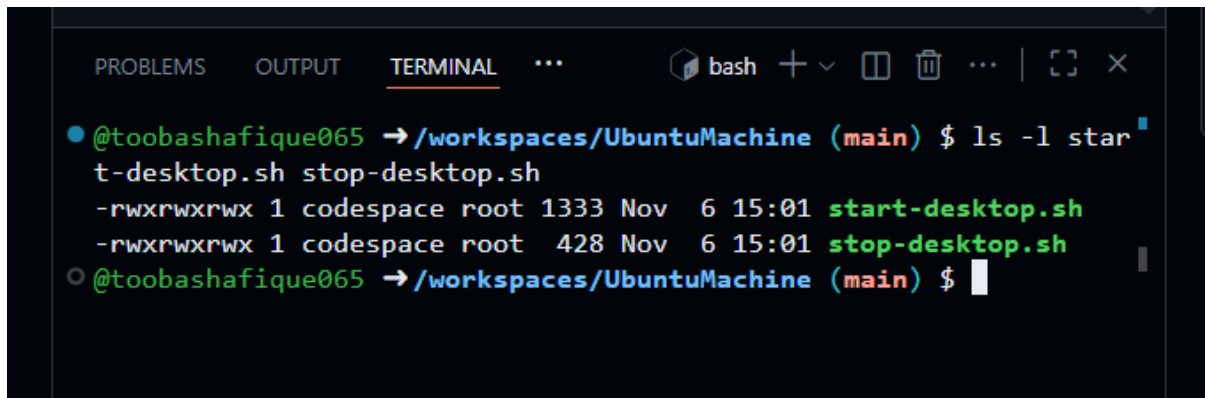


Step 3 – Verify start/stop scripts exist

- Run:
 - `ls -l start-desktop.sh stop-desktop.sh`
 - If needed: `chmod +x start-desktop.sh stop-desktop.sh`

Screenshot

- task14_start_script_ls.png



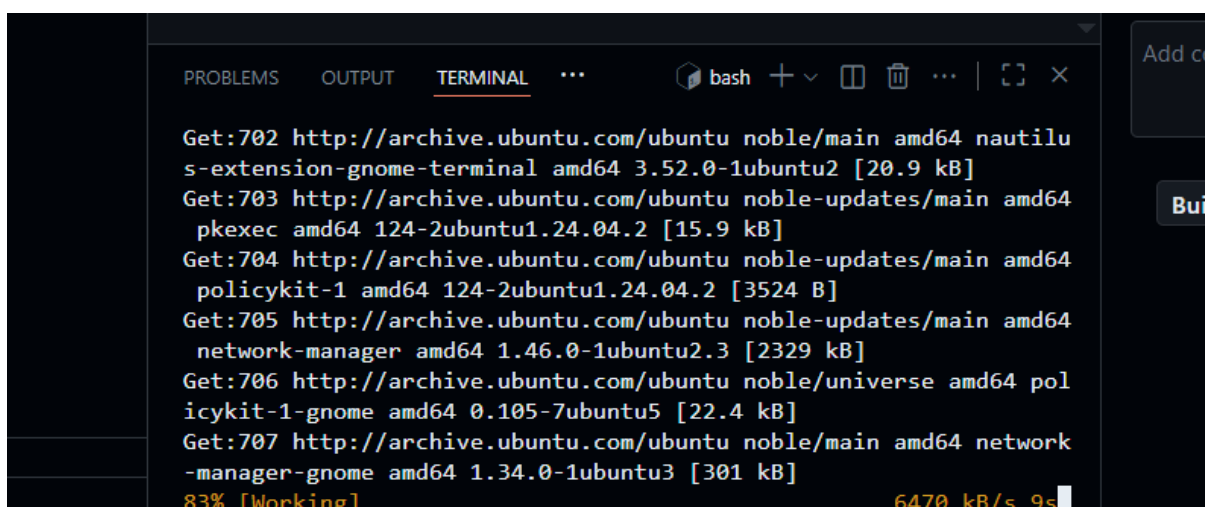
```
PROBLEMS OUTPUT TERMINAL ... bash + v [ ] [ ] ... | [ ] x
@toobashafique065 → /workspaces/UbuntuMachine (main) $ ls -l start-desktop.sh stop-desktop.sh
-rwxrwxrwx 1 codespace root 1333 Nov  6 15:01 start-desktop.sh
-rwxrwxrwx 1 codespace root  428 Nov  6 15:01 stop-desktop.sh
@toobashafique065 → /workspaces/UbuntuMachine (main) $
```

Step 4 – Run start script

- Run:
 - `chmod +x start-desktop.sh`
 - `./start-desktop.sh`

Screenshot

- task14_start_run.png



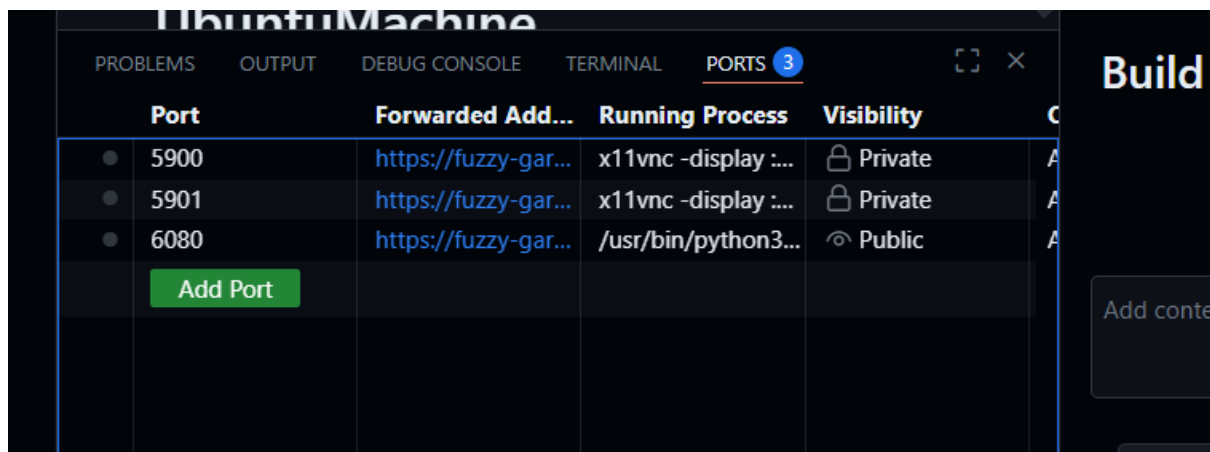
```
PROBLEMS OUTPUT TERMINAL ... bash + v [ ] [ ] ... | [ ] x
Get:702 http://archive.ubuntu.com/ubuntu noble/main amd64 nautilus-extension-gnome-terminal amd64 3.52.0-1ubuntu2 [20.9 kB]
Get:703 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 pkexec amd64 124-2ubuntu1.24.04.2 [15.9 kB]
Get:704 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 policykit-1 amd64 124-2ubuntu1.24.04.2 [3524 B]
Get:705 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 network-manager amd64 1.46.0-1ubuntu2.3 [2329 kB]
Get:706 http://archive.ubuntu.com/ubuntu noble/universe amd64 policykit-1-gnome amd64 0.105-7ubuntu5 [22.4 kB]
Get:707 http://archive.ubuntu.com/ubuntu noble/main amd64 network-manager-gnome amd64 1.34.0-1ubuntu3 [301 kB]
83% [Working] 6470 kB/s 9s
```

Step 5 – Check forwarded ports

- Open Codespaces “Ports” panel.
- Confirm port 6080 is visible and forwarded.

Screenshot

- task14_ports_view.png



Step 6 – Open forwarded port URL and connect to VNC

- Open port 6080 URL in browser.
- Open vnc.html.
- At password prompt enter codespace.

Screenshots

- task14_vnc_url.png

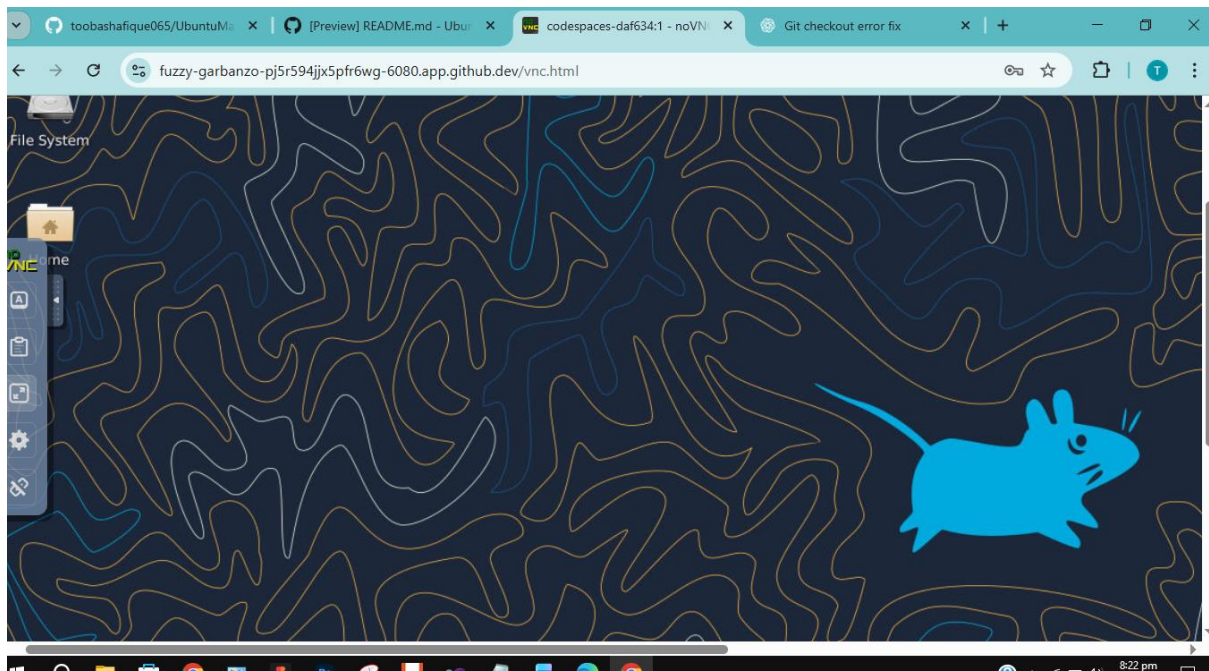
Directory listing for /

- [app/](#)
 - [core/](#)
 - [include/](#)
 - [utils/](#)
 - [vendor/](#)
 - [vnc.html](#)
 - [vnc_auto.html@](#)
 - [vnc_lite.html](#)
-

- (Optional) task14_vnc_connect.png



- task14_vnc_desktop.png



Step 7 – Stop GUI

- In Codespace terminal run: `./stop-desktop.sh`

Screenshot

- `task14_stop_run.png`

```
@toobashafique065 → /workspaces/UbuntuMachine (main) $ ./stop-desktop.sh
[*] Stopping noVNC server...
noVNC server not running.
[*] Stopping x11vnc server...
x11vnc server not running.
[*] Stopping XFCE desktop session...
XFCE4 session not running.
[*] Stopping virtual X server (Xvfb)...
Xvfb not running.
[✓] All services stopped.
@toobashafique065 → /workspaces/UbuntuMachine (main) $
```

Exam Evaluation Questions

Each question below includes the scenario, steps, and required screenshot names as evidence.

Q1. Group Management and Membership

Scenario:

Create groups and manage a user's primary and supplementary group memberships.

Steps:

1. Create groups g1, g2, and g3.

Screenshot: Q1_groups_created.png

```
tooba65@ubuntu:~$ sudo groupadd g1
[sudo] password for tooba65:
tooba65@ubuntu:~$ sudo groupadd g2
tooba65@ubuntu:~$ sudo groupadd g3
```

2. Change examuser's primary group to g3 and add g1 and g2 as supplementary groups.

Screenshot: Q1_group_changes.png

```
tooba65@ubuntu:~$ sudo usermod -g g3 examuser
tooba65@ubuntu:~$ usermod -aG g1,g2 examuser
usermod: Permission denied.
usermod: cannot lock /etc/passwd; try again later.
tooba65@ubuntu:~$ sudo usermod -aG g1,g2 examuser
tooba65@ubuntu:~$
```

3. Show final id command output and /etc/group entries to verify changes.

Screenshot: Q1_group_verification.png

```
tooba65@ubuntu:~$ id examuser
uid=1009(examuser) gid=1008(g3) groups=1008(g3),100(users),1006(g1),1007(g2)
tooba65@ubuntu:~$ grep examuser /etc/group
users:x:100:student,examuser
g1:x:1006:examuser
g2:x:1007:examuser
examuser:x:1009:
tooba65@ubuntu:~$
```

Q2. Ownership and Permission Tasks

Scenario:

Demonstrate ownership changes and apply symbolic and numeric permission changes.

Steps:

1. Create workspace/secret.txt, change its owner to examuser and group to g1.

Screenshot: Q2_chown_chgrp.png

```
tooba65@ubuntu:~$ pwd
/home/tooba65
tooba65@ubuntu:~$ mkdir workspace
tooba65@ubuntu:~$ cd workspace
tooba65@ubuntu:~/workspace$ touch secret.txt
tooba65@ubuntu:~/workspace$ sudo chown examuser:g1 secret.txt
tooba65@ubuntu:~/workspace$ ls -l secret.txt
-rw-rw-r-- 1 examuser g1 0 Nov  5 18:04 secret.txt
tooba65@ubuntu:~/workspace$
```

2. Remove all permissions for group and others using symbolic command, then use numeric command to achieve the same result.

Screenshot: Q2_symbolic_numeric.png

```
tooba65@ubuntu:~/workspace$ ls -l secret.txt
-rw-rw-r-- 1 examuser g1 0 Nov  5 18:04 secret.txt
tooba65@ubuntu:~/workspace$ chmod go-rwx secret.txt
chmod: changing permissions of 'secret.txt': Operation not permitted
tooba65@ubuntu:~/workspace$ sudo chmod go-rwx secret.txt
tooba65@ubuntu:~/workspace$ ls -l secret.txt
-rw----- 1 examuser g1 0 Nov  5 18:04 secret.txt
tooba65@ubuntu:~/workspace$
```

ent examples

3. Show ls -l output for the file after each change to confirm permissions.

Screenshot: Q2_permissions_ls.png

```
tooba65@ubuntu:~/workspace$ sudo chown examuser:g1 secret.txt
tooba65@ubuntu:~/workspace$ ls -l secret.txt
-rw-rw-r-- 1 examuser g1 0 Nov  5 18:04 secret.txt
tooba65@ubuntu:~/workspace$ chmod go-rwx secret.txt
chmod: changing permissions of 'secret.txt': Operation not permitted
tooba65@ubuntu:~/workspace$ sudo chmod go-rwx secret.txt
tooba65@ubuntu:~/workspace$ ls -l secret.txt
-rw----- 1 examuser g1 0 Nov  5 18:04 secret.txt
tooba65@ubuntu:~/workspace$ sudo chmod 600 secret.txt
tooba65@ubuntu:~/workspace$ ls -l secret.txt
-rw----- 1 examuser g1 0 Nov  5 18:04 secret.txt
tooba65@ubuntu:~/workspace$
```

Q3. Pipes, Grep, and Redirection Practice

Scenario:

Filter system logs and save results using redirection and piping.

Steps:

1. Use grep (or journalctl) with a pipe to find lines containing “error” or “fail” and display first 20 results.

Screenshot: Q3_grep_pipe.png

```
tooba65@ubuntu:~$ sudo grep -Ei "error|fail" /var/log/syslog | head -20
[sudo] password for tooba65:
2025-11-02T11:00:36.419258+00:00 ubuntu multipathd[480]: sda: failed to get udev uid: No data available
2025-11-02T11:00:36.419540+00:00 ubuntu multipath: sda: failed to get sysfs uid: No such file or directory
2025-11-02T11:00:36.419548+00:00 ubuntu multipath: sda: failed to get sgio uid: No such file or directory
2025-11-02T11:00:36.419616+00:00 ubuntu multipathd[480]: sda: failed to get udev uid: No data available
2025-11-02T11:00:36.419623+00:00 ubuntu multipathd[480]: sda: failed to get path uid
2025-11-02T11:00:36.419630+00:00 ubuntu multipathd[480]: uevent trigger error
2025-11-02T11:00:36.420999+00:00 ubuntu systemd[1]: apport-autoreport.path - Process error reports when automatic report
ing is enabled (file watch) was skipped because of an unmet condition check (ConditionPathExists=/var/lib/apport/autorep
ort).
2025-11-02T11:00:36.421066+00:00 ubuntu systemd[1]: apport-autoreport.timer - Process error reports when automatic repor
ting is enabled (timer based) was skipped because of an unmet condition check (ConditionPathExists=/var/lib/apport/autor
eport).
2025-11-02T11:00:36.423032+00:00 ubuntu systemd[1]: Starting grub-initrd-fallback.service - GRUB failed boot detection..
.
2025-11-02T11:00:36.423202+00:00 ubuntu systemd[1]: rsyslog.service: Failed with result 'protocol'.
2025-11-02T11:00:36.423256+00:00 ubuntu systemd[1]: Failed to start rsyslog.service - System Logging Service.
2025-11-02T11:00:36.423410+00:00 ubuntu systemd[1]: Finished grub-initrd-fallback.service - GRUB failed boot detection.
2025-11-02T11:00:36.458898+00:00 ubuntu kernel: ACPI: _OSC evaluation for CPUs failed, trying _PDC
2025-11-02T11:00:36.468839+00:00 ubuntu kernel: pci 0000:00:15.3: bridge window [io size 0x1000]: failed to assign
2025-11-02T11:00:36.468841+00:00 ubuntu kernel: pci 0000:00:15.4: bridge window [io size 0x1000]: failed to assign
2025-11-02T11:00:36.468842+00:00 ubuntu kernel: pci 0000:00:15.5: bridge window [io size 0x1000]: failed to assign
2025-11-02T11:00:36.468844+00:00 ubuntu kernel: pci 0000:00:15.6: bridge window [io size 0x1000]: failed to assign
2025-11-02T11:00:36.468845+00:00 ubuntu kernel: pci 0000:00:15.7: bridge window [io size 0x1000]: failed to assign
2025-11-02T11:00:36.468846+00:00 ubuntu kernel: pci 0000:00:16.3: bridge window [io size 0x1000]: failed to assign
2025-11-02T11:00:36.468848+00:00 ubuntu kernel: pci 0000:00:16.4: bridge window [io size 0x1000]: failed to assign
tooba65@ubuntu:~$
```

2. Save filtered results to ~/logs/errors.txt (overwrite), then append more results using append redirection.

Screenshot: Q3_redirect_overwrite_append.png

```
tooba65@ubuntu:~$ mkdir -p ~/logs
tooba65@ubuntu:~$ sudo grep -Ei "error|fail" /var/log/syslog > ~/logs/errors.txt
tooba65@ubuntu:~$ cat ~/logs/errors.txt
2025-11-02T11:00:36.419258+00:00 ubuntu multipathd[480]: sda: failed to get udev uid: No data available
2025-11-02T11:00:36.419540+00:00 ubuntu multipath: sda: failed to get sysfs uid: No such file or directory
2025-11-02T11:00:36.419548+00:00 ubuntu multipath: sda: failed to get sgio uid: No such file or directory
2025-11-02T11:00:36.419616+00:00 ubuntu multipathd[480]: sda: failed to get udev uid: No data available
2025-11-02T11:00:36.419623+00:00 ubuntu multipathd[480]: sda: failed to get path uid
2025-11-02T11:00:36.419630+00:00 ubuntu multipathd[480]: uevent trigger error
2025-11-02T11:00:36.420999+00:00 ubuntu systemd[1]: appport-autoreport.path - Process error reports when automatic reporting is enabled (file watch) was skipped because of an unmet condition check (ConditionPathExists=/var/lib/appport/autoreport).
2025-11-02T11:00:36.421066+00:00 ubuntu systemd[1]: appport-autoreport.timer - Process error reports when automatic reporting is enabled (timer based) was skipped because of an unmet condition check (ConditionPathExists=/var/lib/appport/autoreport).
2025-11-02T11:00:36.423032+00:00 ubuntu systemd[1]: Starting grub-initrd-fallback.service - GRUB failed boot detection.
.
2025-11-02T11:00:36.423202+00:00 ubuntu systemd[1]: rsyslog.service: Failed with result 'protocol'.
2025-11-02T11:00:36.423256+00:00 ubuntu systemd[1]: Failed to start rsyslog.service - System Logging Service.
2025-11-02T11:00:36.423410+00:00 ubuntu systemd[1]: Finished grub-initrd-fallback.service - GRUB failed boot detection.
2025-11-02T11:00:36.458898+00:00 ubuntu kernel: ACPI: _OSC evaluation for CPUs failed, trying _PDC
2025-11-02T11:00:36.468839+00:00 ubuntu kernel: pci 0000:00:15.3: bridge window [io size 0x1000]: failed to assign
2025-11-02T11:00:36.468841+00:00 ubuntu kernel: pci 0000:00:15.4: bridge window [io size 0x1000]: failed to assign
2025-11-02T11:00:36.468842+00:00 ubuntu kernel: pci 0000:00:15.5: bridge window [io size 0x1000]: failed to assign
2025-11-02T11:00:36.468844+00:00 ubuntu kernel: pci 0000:00:15.6: bridge window [io size 0x1000]: failed to assign
```

3. View the saved file using a pager less view of saved file

Screenshot: Q3_pager_view.png

```
tooba65@ubuntu:~$ cat ~/logs/errors.txt | less
2025-11-02T11:00:36.419258+00:00 ubuntu multipathd[480]: sda: failed to get udev uid: No data available
2025-11-02T11:00:36.419540+00:00 ubuntu multipath: sda: failed to get sysfs uid: No such file or directory
2025-11-02T11:00:36.419548+00:00 ubuntu multipath: sda: failed to get sgio uid: No such file or directory
2025-11-02T11:00:36.419616+00:00 ubuntu multipathd[480]: sda: failed to get udev uid: No data available
2025-11-02T11:00:36.419623+00:00 ubuntu multipathd[480]: sda: failed to get path uid
2025-11-02T11:00:36.419630+00:00 ubuntu multipathd[480]: uevent trigger error
2025-11-02T11:00:36.420999+00:00 ubuntu systemd[1]: appport-autoreport.path - Process error reports when automatic reporting is enabled (file watch) was skipped because of an unmet condition check (ConditionPathExists=/var/lib/appport/autoreport).
2025-11-02T11:00:36.421066+00:00 ubuntu systemd[1]: appport-autoreport.timer - Process error reports when automatic reporting is enabled (timer based) was skipped because of an unmet condition check (ConditionPathExists=/var/lib/appport/autoreport).
2025-11-02T11:00:36.423032+00:00 ubuntu systemd[1]: Starting grub-initrd-fallback.service - GRUB failed boot detection..
.
2025-11-02T11:00:36.423202+00:00 ubuntu systemd[1]: rsyslog.service: Failed with result 'protocol'.
2025-11-02T11:00:36.423256+00:00 ubuntu systemd[1]: Failed to start rsyslog.service - System Logging Service.
2025-11-02T11:00:36.423410+00:00 ubuntu systemd[1]: Finished grub-initrd-fallback.service - GRUB failed boot detection.
2025-11-02T11:00:36.458898+00:00 ubuntu kernel: ACPI: _OSC evaluation for CPUs failed, trying _PDC
2025-11-02T11:00:36.468839+00:00 ubuntu kernel: pci 0000:00:15.3: bridge window [io size 0x1000]: failed to assign
2025-11-02T11:00:36.468841+00:00 ubuntu kernel: pci 0000:00:15.4: bridge window [io size 0x1000]: failed to assign
2025-11-02T11:00:36.468842+00:00 ubuntu kernel: pci 0000:00:15.5: bridge window [io size 0x1000]: failed to assign
2025-11-02T11:00:36.468844+00:00 ubuntu kernel: pci 0000:00:15.6: bridge window [io size 0x1000]: failed to assign
2025-11-02T11:00:36.468845+00:00 ubuntu kernel: pci 0000:00:15.7: bridge window [io size 0x1000]: failed to assign
2025-11-02T11:00:36.468846+00:00 ubuntu kernel: pci 0000:00:16.3: bridge window [io size 0x1000]: failed to assign
2025-11-02T11:00:36.468848+00:00 ubuntu kernel: pci 0000:00:16.4: bridge window [io size 0x1000]: failed to assign
2025-11-02T11:00:36.468849+00:00 ubuntu kernel: pci 0000:00:16.5: bridge window [io size 0x1000]: failed to assign
2025-11-02T11:00:36.468850+00:00 ubuntu kernel: pci 0000:00:16.6: bridge window [io size 0x1000]: failed to assign
2025-11-02T11:00:36.468851+00:00 ubuntu kernel: pci 0000:00:16.7: bridge window [io size 0x1000]: failed to assign
2025-11-02T11:00:36.468853+00:00 ubuntu kernel: pci 0000:00:17.3: bridge window [io size 0x1000]: failed to assign
/home/tooba65/logs/errors.txt
```

Q4. Script: Variables, Command Substitution, File & Dir Checks

Scenario:

Build and run a script showing variables, command substitution, and filesystem checks.

Steps:

1. Create setup.sh with a shebang and a variable var1 that you echo.

Screenshot: Q4_step1_var1.png


```
tooba65@ubuntu: ~  
#!/bin/bash  
var1="Hello from Exam Evaluation"  
echo "var1: $var1"  
~  
~  
~  
~  
tooba65@ubuntu:~$ ./setup.sh  
var1: Hello from Exam Evaluation  
tooba65@ubuntu:~$
```

2. Append command substitution to store ls -l output in a variable and echo it.

Screenshot: Q4_step2_allfiles.png

```
tooba65@ubuntu: ~  
#!/bin/bash  
var1="Hello from Exam Evaluation"  
echo "var1: $var1"  
# Step 2: Command substitution  
allfiles=$(ls -l)  
echo "All files in current directory:"  
echo "$allfiles"
```

```
tooba65@ubuntu:~$ vim setup.sh  
tooba65@ubuntu:~$ tooba65@ubuntu:~$ ./setup.sh  
var1: Hello from Exam Evaluation  
All files in current directory:  
total 411988  
-rw-rw-r-- 1 tooba65 tooba65      177 Oct 17 05:43 answers.md  
-rw-rw-r-- 1 tooba65 tooba65      461 Oct 24 05:17 apt_update_vs_upgrade.md  
drwxr-xr-x 2 tooba65 tooba65    4096 Oct 24 07:02 Desktop  
-rw-rw-r-- 1 tooba65 tooba65 421490404 Oct 30 11:38 docker-desktop-amd64.deb  
drwxr-xr-x 2 tooba65 tooba65    4096 Oct 24 07:02 Documents  
drwxr-xr-x 2 tooba65 tooba65    4096 Nov  2 13:50 Downloads  
drwxrwxr-x 3 tooba65 tooba65    4096 Oct 21 14:21 forensic_lab  
drwxrwxr-x 3 tooba65 tooba65    4096 Oct 21 14:32 forensic_lab_backup  
drwxrwxr-x 3 tooba65 tooba65    4096 Oct 17 05:59 lab4  
drwxrwxr-x 2 tooba65 tooba65    4096 Oct 24 18:17 Lab5  
drwxrwxr-x 2 tooba65 tooba65    4096 Nov  6 11:04 logs  
drwxr-xr-x 2 tooba65 tooba65    4096 Oct 24 07:02 Music  
drwxr-xr-x 2 tooba65 tooba65    4096 Oct 24 07:02 Pictures  
drwxr-xr-x 2 tooba65 tooba65    4096 Oct 24 07:02 Public  
-rw-rw-r-- 1 tooba65 tooba65      147 Oct 21 14:19 Q2_report.md  
-rwxrwxr-x 1 tooba65 tooba65      176 Nov  6 11:14 setup.sh  
drwx----- 3 tooba65 tooba65    4096 Oct 24 05:58 snap  
-rw-rw-r-- 1 tooba65 tooba65 298809 Nov  5 15:08 syslog_systemd.txt  
drwxr-xr-x 2 tooba65 tooba65    4096 Oct 24 07:02 Templates  
drwxrwxr-t 2 tooba65 tooba65    4096 Oct 24 07:02 thinclient_drives  
drwxr-xr-x 2 tooba65 tooba65    4096 Oct 24 07:02 Videos  
drwxrwxr-x 2 tooba65 tooba65    4096 Nov  5 18:04 workspace  
tooba65@ubuntu:~$
```

3. Append directory and file checks to create dir1 and dir1/file2 if missing, and show final permissions.

Screenshot: Q4_step3_dirfile_checks.png

```

tooba65@ubuntu: ~
var1="Hello from Exam Evaluation"
echo "var1: $var1"
# Step 2: Command substitution
allfiles=$(ls -l)
echo "All files in current directory:"
echo "$allfiles"
# Step 3: Directory and file checks
if [ ! -d dir1 ]; then
    mkdir dir1
    echo "Directory dir1 created."
else
    echo "Directory dir1 already exists."
fi
if [ ! -f dir1/file2 ]; then
    touch dir1/file2
    echo "File dir1/file2 created."
else
    echo "File dir1/file2 already exists."
fi
echo "Final permissions for dir1 and file2:"
ls -ld dir1
ls -l dir1/file2

```

```

tooba65@ubuntu:~$ tooba65@ubuntu:~$ ./setup.sh
var1: Hello from Exam Evaluation
All files in current directory:
total 411988
-rw-rw-r-- 1 tooba65 tooba65      177 Oct 17 05:43 answers.md
-rw-rw-r-- 1 tooba65 tooba65      461 Oct 24 05:17 apt_update_vs_upgrade.md
drwxr-xr-x 2 tooba65 tooba65    4096 Oct 24 07:02 Desktop
-rw-rw-r-- 1 tooba65 tooba65 421490404 Oct 30 11:38 docker-desktop-amd64.deb
drwxr-xr-x 2 tooba65 tooba65    4096 Oct 24 07:02 Documents
drwxr-xr-x 2 tooba65 tooba65    4096 Nov  2 13:50 Downloads
drwxrwxr-x 3 tooba65 tooba65    4096 Oct 21 14:21 forensic_lab
drwxrwxr-x 3 tooba65 tooba65    4096 Oct 21 14:32 forensic_lab_backup
drwxrwxr-x 3 tooba65 tooba65    4096 Oct 17 05:59 lab4
drwxrwxr-x 2 tooba65 tooba65    4096 Oct 24 18:17 Lab5
drwxrwxr-x 2 tooba65 tooba65    4096 Nov  6 11:04 logs
drwxr-xr-x 2 tooba65 tooba65    4096 Oct 24 07:02 Music
drwxr-xr-x 2 tooba65 tooba65    4096 Oct 24 07:02 Pictures
drwxr-xr-x 2 tooba65 tooba65    4096 Oct 24 07:02 Public
-rw-rw-r-- 1 tooba65 tooba65     147 Oct 21 14:19 Q2_report.md
-rwxrwxr-x 1 tooba65 tooba65     578 Nov  6 11:16 setup.sh
drwx----- 3 tooba65 tooba65    4096 Oct 24 05:58 snap
-rw-rw-r-- 1 tooba65 tooba65 298809 Nov  5 15:08 syslog_systemd.txt
drwxr-xr-x 2 tooba65 tooba65    4096 Oct 24 07:02 Templates
drwxrwxr-t 2 tooba65 tooba65    4096 Oct 24 07:02 thinclient_drives
drwxr-xr-x 2 tooba65 tooba65    4096 Oct 24 07:02 Videos
drwxrwxr-x 2 tooba65 tooba65    4096 Nov  5 18:04 workspace
Directory dir1 created.
File dir1/file2 created.
Final permissions for dir1 and file2:
drwxrwxr-x 2 tooba65 tooba65 4096 Nov  6 11:16 dir1
-rw-rw-r-- 1 tooba65 tooba65 0 Nov  6 11:16 dir1/file2
tooba65@ubuntu:~$

```

Q5. Script: Comparisons and String Tests

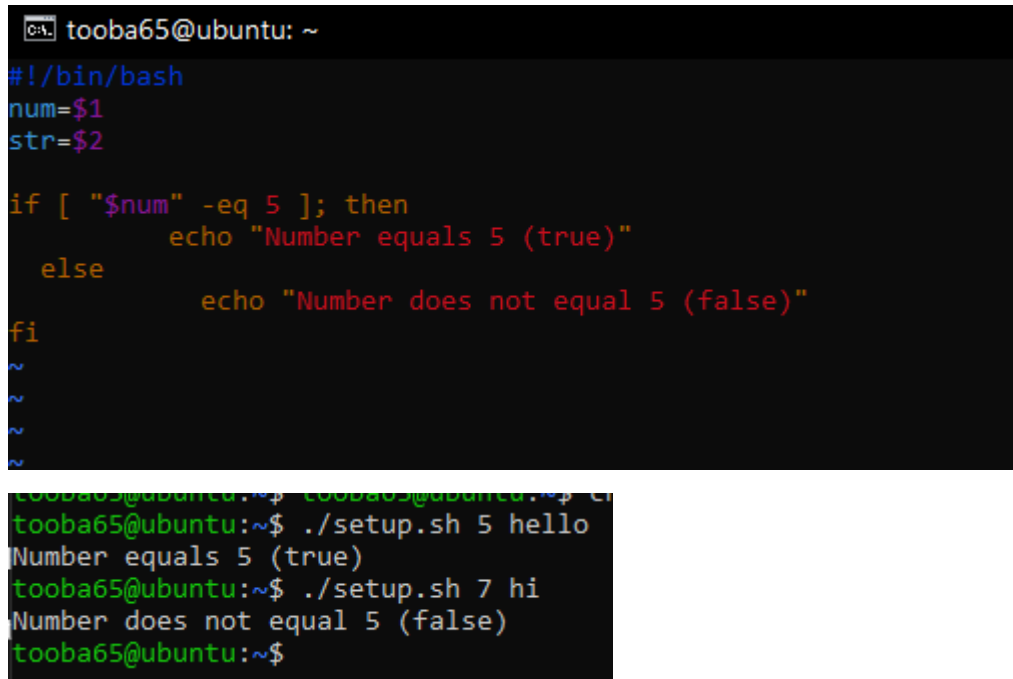
Scenario:

Add numeric and string comparison tests to a script and show both true/false cases.

Steps:

1. Overwrite setup.sh to set num=\$1 and str=\$2, add an -eq test with true and false cases.

Screenshot: Q5_eq_examples.png



```
tooba65@ubuntu: ~  
#!/bin/bash  
num=$1  
str=$2  
  
if [ "$num" -eq 5 ]; then  
    echo "Number equals 5 (true)"  
else  
    echo "Number does not equal 5 (false)"  
fi  
~  
~  
~  
~  
tooba65@ubuntu:~$ ./setup.sh 5 hello  
Number equals 5 (true)  
tooba65@ubuntu:~$ ./setup.sh 7 hi  
Number does not equal 5 (false)  
tooba65@ubuntu:~$
```

2. Append numeric comparison tests: -ne, -gt, -lt, -ge, -le, showing true and false examples for each.

Screenshot: Q5_numeric_tests.png

```

tooba65@ubuntu: ~
if [ "$num" -eq 5 ]; then
    echo "Number equals 5 (true)"
else
    echo "Number does not equal 5 (false)"
fi
# Step 2: Other numeric tests
if [ "$num" -ne 5 ]; then
    echo "$num is not equal to 5 (-ne test true)"
else
    echo "$num is equal to 5 (-ne test false)"
fi
if [ "$num" -gt 3 ]; then
    echo "$num is greater than 3 (-gt test true)"
else
    echo "$num is not greater than 3 (-gt test false)"
fi
if [ "$num" -lt 10 ]; then
    echo "$num is less than 10 (-lt test true)"
else
    echo "$num is not less than 10 (-lt test false)"
fi
if [ "$num" -ge 5 ]; then
    echo "$num is greater than or equal to 5 (-ge test true)"
else
    echo "$num is less than 5 (-ge test false)"
fi
if [ "$num" -le 5 ]; then
    echo "$num is less than or equal to 5 (-le test true)"
else
    echo "$num is greater than 5 (-le test false)"
fi

```

```

tooba65@ubuntu: ~$ ./setup.sh 5 hi
Number equals 5 (true)
5 is equal to 5 (-ne test false)
5 is greater than 3 (-gt test true)
5 is less than 10 (-lt test true)
5 is greater than or equal to 5 (-ge test true)
5 is less than or equal to 5 (-le test true)
tooba65@ubuntu: ~$ ./setup.sh 8 hi
Number does not equal 5 (false)
8 is not equal to 5 (-ne test true)
8 is greater than 3 (-gt test true)
8 is less than 10 (-lt test true)
8 is greater than or equal to 5 (-ge test true)
8 is greater than 5 (-le test false)
tooba65@ubuntu: ~$

```

3. Append string equality (=), inequality (!=), and zero-length (-z) tests for \$2, showing both outcomes.

Screenshot: Q5_string_tests.png


```

    echo "$num is less than 5 (-lt test true)"
fi
if [ "$num" -le 5 ]; then
    echo "$num is less than or equal to 5 (-le test true)"
else
    echo "$num is greater than 5 (-le test false)"
fi
# Step 3: String tests
if [ "$str" = "hello" ]; then
    echo "String equals 'hello' (= test true)"
else
    echo "String does not equal 'hello' (= test false)"
fi
if [ "$str" != "world" ]; then
    echo "String is not 'world' (!= test true)"
else
    echo "String is 'world' (!= test false)"
fi
if [ -z "$str" ]; then
    echo "String is empty (-z test true)"
else
    echo "String is not empty (-z test false)"
fi
-- INSERT --

```

```

tooba65@ubuntu:~$ tooba65@ubuntu:~$ ./setup.sh 5 hello
Number equals 5 (true)
5 is equal to 5 (-ne test false)
5 is greater than 3 (-gt test true)
5 is less than 10 (-lt test true)
5 is greater than or equal to 5 (-ge test true)
5 is less than or equal to 5 (-le test true)
String equals 'hello' (= test true)
String is not 'world' (!= test true)
String is not empty (-z test false)
tooba65@ubuntu:~$

```

Q6. Script: For Loop and Argument Handling

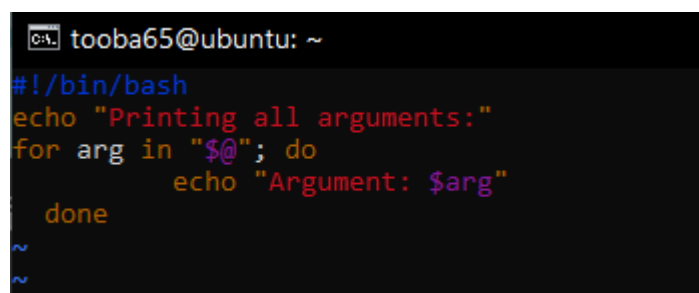
Scenario:

Write a script that prints all provided arguments and handles quoted multi-word arguments.

Steps:

1. Create or overwrite setup.sh to print every argument using "\$@" in a for loop.

Screenshot: Q6_script_forloop_vim.png



```

tooba65@ubuntu: ~
#!/bin/bash
echo "Printing all arguments:"
for arg in "$@"; do
    echo "Argument: $arg"
done
~
~

```

2. Run the script with single and quoted multi-word arguments, each printed on its own line.

Screenshot: Q6_forloop_run.png

```
tooba65@ubuntu:~$ tooba65@ubuntu:~$ vim setup.sh
tooba65@ubuntu:~$ tooba65@ubuntu:~$ chmod +x setup.sh
tooba65@ubuntu:~$ ./setup.sh one "two words" three
Printing all arguments:
Argument: one
Argument: two words
Argument: three
tooba65@ubuntu:~$
```

Q7. Script: While Loop Summation and Functions

Scenario:

Implement a summation function and demonstrate function-based arithmetic.

Steps:

1. Write an interactive while-loop that accumulates numbers until "q" is entered, showing running totals.

Screenshot: Q7_while_session.png

```
tooba65@ubuntu: ~
#!/bin/bash
~
~
~
~
~
```

```
tooba65@ubuntu:~$ vim setup.sh
tooba65@ubuntu:~$ tooba65@ubuntu:~$ chmod +x setup.sh
tooba65@ubuntu:~$ ./setup.sh
tooba65@ubuntu:~$
```

```
tooba65@ubuntu: ~
#!/bin/bash
# Interactive while-loop summation
sum=0
while true; do
    read -p "Enter a number (or 'q' to quit): " input
    if [ "$input" = "q" ]; then
        break
    fi
    sum=$((sum + input))
    echo "Total Score: $sum"
done
echo "Final total: $sum"
~
~
```

```

tooba65@ubuntu:~$ vim setup.sh
tooba65@ubuntu:~$ tooba65@ubuntu:~$ ./setup.sh
Enter a number (or 'q' to quit): 5
Total Score: 5
Enter a number (or 'q' to quit): 3
Total Score: 8
Enter a number (or 'q' to quit): 2
Total Score: 10
Enter a number (or 'q' to quit): q
Final total: 10
tooba65@ubuntu:~$

```

2. Add a function that accepts two numbers, returns their sum, and show the result stored in a variable.

Screenshot: Q7_function_sum.png

```

#!/bin/bash
# Interactive while-loop summation
sum=0
while true; do
    read -p "Enter a number (or 'q' to quit): " input
    if [ "$input" = "q" ]; then
        break
    fi
    sum=$((sum + input))
    echo "Total Score: $sum"
done
echo "Final total: $sum"
# Function that sums two numeric arguments
sum_args() {
    a=$1
    b=$2
    echo $((a + b))
}
# Demonstration of sum_args
result=$(sum_args 4 9)
echo "sum_args(4,9) returned: $result"

```

```

tooba65@ubuntu:~$ vim setup.sh
tooba65@ubuntu:~$ tooba65@ubuntu:~$ ./setup.sh
Enter a number (or 'q' to quit): 4
Total Score: 4
Enter a number (or 'q' to quit): 5
Total Score: 9
Enter a number (or 'q' to quit): q
Final total: 9
sum_args(4,9) returned: 13
tooba65@ubuntu:~$

```